which says that if f is integrated and then the result is differentiated, we arrive back at the original function f. And we reformulated Part 2 as the Net Change Theorem:

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

This version says that if we take a function F, first differentiate it, and then integrate the result, we arrive back at the original function F, but in the form $F(b) - F(a)$. Taken together, the two parts of the Fundamental Theorem of Calculus say that differentiation and integration are inverse processes. Each undoes what the other does.

The Fundamental Theorem of Calculus is unquestionably the most important theorem in calculus and, indeed, it ranks as one of the great accomplishments of the human mind. Before it was discovered, from the time of Eudoxus and Archimedes to the time of Galileo and Fermat, problems of finding areas, volumes, and lengths of curves were so difficult that only a genius could meet the challenge. But now, armed with the systematic method that Newton and Leibniz fashioned out of the Fundamental Theorem, we will see in the chapters to come that these challenging problems are accessible to all of us.

EXERCISES 5.3

1–28 Evaluate the integral.

1. $\displaystyle\int_{-2}^{3} (x^2 - 3)\,dx$

2. $\displaystyle\int_{1}^{2} x^{-2}\,dx$

3. $\displaystyle\int_{0}^{2} \left(x^4 - \tfrac{3}{4}x^2 + \tfrac{2}{3}x - 1\right)dx$

4. $\displaystyle\int_{0}^{1} \left(1 + \tfrac{1}{2}u^4 - \tfrac{2}{5}u^9\right)du$

5. $\displaystyle\int_{0}^{1} x^{4/5}\,dx$

6. $\displaystyle\int_{1}^{8} \sqrt[3]{x}\,dx$

7. $\displaystyle\int_{-1}^{0} (2x - e^x)\,dx$

8. $\displaystyle\int_{-5}^{5} e\,dx$

9. $\displaystyle\int_{1}^{2} (1 + 2y)^2\,dy$

10. $\displaystyle\int_{0}^{2} (y - 1)(2y + 1)\,dy$

11. $\displaystyle\int_{1}^{9} \frac{x - 1}{\sqrt{x}}\,dx$

12. $\displaystyle\int_{-1}^{1} t(1 - t)^2\,dt$

13. $\displaystyle\int_{0}^{1} x\left(\sqrt[3]{x} + \sqrt[4]{x}\right)dx$

14. $\displaystyle\int_{0}^{\pi/4} \sec\theta\,\tan\theta\,d\theta$

15. $\displaystyle\int_{0}^{\pi/4} \sec^2 t\,dt$

16. $\displaystyle\int_{1}^{18} \sqrt{\frac{3}{z}}\,dz$

17. $\displaystyle\int_{1}^{9} \frac{1}{2x}\,dx$

18. $\displaystyle\int_{0}^{5} (2e^x + 4\cos x)\,dx$

19. $\displaystyle\int_{0}^{1} (x^e + e^x)\,dx$

20. $\displaystyle\int_{0}^{1} 10^x\,dx$

21. $\displaystyle\int_{-1}^{1} e^{u+1}\,du$

22. $\displaystyle\int_{0}^{1} \frac{4}{t^2 + 1}\,dt$

23. $\displaystyle\int_{1}^{2} \frac{v^3 + 3v^6}{v^4}\,dv$

24. $\displaystyle\int_{0}^{\pi/3} \frac{\sin\theta + \sin\theta\,\tan^2\theta}{\sec^2\theta}\,d\theta$

25. $\displaystyle\int_{0}^{\pi/4} \frac{1 + \cos^2\theta}{\cos^2\theta}\,d\theta$

26. $\displaystyle\int_{1}^{2} \frac{(x - 1)^3}{x^2}\,dx$

27. $\displaystyle\int_{0}^{1/\sqrt{3}} \frac{t^2 - 1}{t^4 - 1}\,dt$

28. $\displaystyle\int_{0}^{2} |2x - 1|\,dx$

29–30 What is wrong with the equation?

29. $\displaystyle\int_{-1}^{3} \frac{1}{x^2}\,dx = \frac{x^{-1}}{-1}\bigg]_{-1}^{3} = -\frac{4}{3}$

30. $\displaystyle\int_{0}^{\pi} \sec^2 x\,dx = \tan x\Big]_{0}^{\pi} = 0$

31–32 Use a graph to give a rough estimate of the area of the region that lies beneath the given curve. Then find the exact area.

31. $y = \sin x,\ 0 \leqslant x \leqslant \pi$

32. $y = \sec^2 x,\ 0 \leqslant x \leqslant \pi/3$

33–34 Evaluate the integral and interpret it as a difference of areas. Illustrate with a sketch.

33. $\displaystyle\int_{-1}^{2} x^3\,dx$

34. $\displaystyle\int_{-\pi/2}^{2\pi} \cos x\,dx$

35–36 Verify by differentiation that the formula is correct.

35. $\displaystyle\int \cos^3 x\,dx = \sin x - \tfrac{1}{3}\sin^3 x + C$

36. $\displaystyle\int x\cos x\,dx = x\sin x + \cos x + C$

37–38 Find the general indefinite integral. Illustrate by graphing several members of the family on the same screen.

37. $\int \left(\cos x + \frac{1}{2}x\right) dx$

38. $\int (e^x - 2x^2)\, dx$

39–44 Find the general indefinite integral.

39. $\int (1 - t)(2 + t^2)\, dt$

40. $\int v(v^2 + 2)^2\, dv$

41. $\int (1 + \tan^2\alpha)\, d\alpha$

42. $\int \sec t\,(\sec t + \tan t)\, dt$

43. $\int \frac{\sin x}{1 - \sin^2 x}\, dx$

44. $\int \frac{\sin 2x}{\sin x}\, dx$

45. Measles pathogenesis The function

$$f(t) = -t(t - 21)(t + 1)$$

has been used to model the measles virus concentration in an infected individual. The area under the graph of f represents the total amount of infection. We saw in Section 5.1 that at $t = 12$ days this total amount of infection reaches the threshold beyond which symptoms appear. Use the Evaluation Theorem to calculate this threshold value.

46. If $V'(t)$ is the rate at which water flows into a reservoir at time t, what does the integral

$$\int_{t_1}^{t_2} V'(t)\, dt$$

represent?

47. Growth rate If $w'(t)$ is the rate of growth of a child in pounds per year, what does $\int_5^{10} w'(t)\, dt$ represent?

48. Age-structured populations Suppose the number of individuals of age a is given by the function $N(a)$ (number of individuals per age a). What does the integral $\int_0^{15} N(a)\, da$ represent?

49. Sea urchins Integration is sometimes used when censusing a population. For example, suppose the density of sea urchins at different points x along a coastline is given by the function $f(x)$ individuals per meter, where x is the distance (in meters) along the coast from the start of the species' range. What does the integral $\int_a^b f(x)\, dx$ represent?

© Vilainecrevette / Shutterstock.com

50. Bacteria growth A bacteria colony increases in size at a rate of $4.0553e^{1.8t}$ bacteria per hour. If the initial population is 46 bacteria, find the population four hours later.

51. In a chemical reaction, the rate of reaction is the derivative of the concentration $[\mathrm{C}](t)$ of the product of the reaction. What does

$$\int_{t_1}^{t_2} \frac{d[\mathrm{C}]}{dt}\, dt$$

represent?

52. A **honeybee population** starts with 100 bees and increases at a rate of $n'(t)$ bees per week. What does the expression $100 + \int_0^{15} n'(t)\, dt$ represent?

53. If oil leaks from a tank at a rate of $r(t)$ gallons per minute at time t, what does $\int_0^{120} r(t)\, dt$ represent?

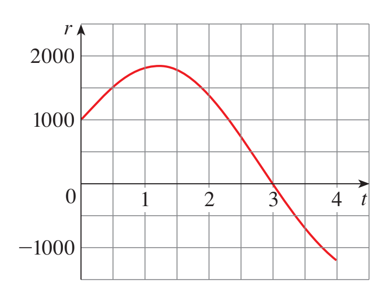
54. Suppose that a volcano is erupting and readings of the rate $r(t)$ at which solid materials are spewed into the atmosphere are given in the table. The time t is measured in seconds and the units for $r(t)$ are tonnes (metric tons) per second.

t	0	1	2	3	4	5	6
$r(t)$	2	10	24	36	46	54	60

(a) Give upper and lower estimates for the total quantity $Q(6)$ of erupted materials after 6 seconds.
(b) Use the Midpoint Rule to estimate $Q(6)$.

55. Water flows from the bottom of a storage tank at a rate of $r(t) = 200 - 4t$ liters per minute, where $0 \leqslant t \leqslant 50$. Find the amount of water that flows from the tank during the first 10 minutes.

56. Water flows into and out of a storage tank. A graph of the rate of change $r(t)$ of the volume of water in the tank, in liters per day, is shown. If the amount of water in the tank at time $t = 0$ is 25,000 L, use the Midpoint Rule to estimate the amount of water in the tank four days later.



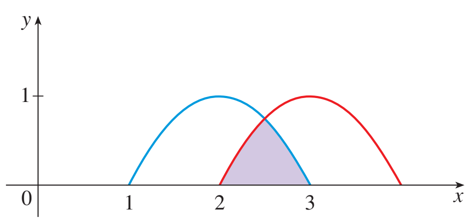
57. Von Bertalanffy growth Many fish grow in a way that is described by the von Bertalanffy growth equation. For a fish that starts life with a length of 1 cm and has a maximum length of 30 cm, this equation predicts that the growth rate is $29e^{-a}$ cm/year, where a is the age of the fish. How long will the fish be after 5 years?

58. Niche overlap The extent to which species compete for resources is often measured by the *niche overlap*. If the horizontal axis represents a continuum of different resource types (for example, seed sizes for certain bird species), then a plot of the degree of preference for these resources is called a *species' niche*. The degree of overlap of two species' niches is then a measure of the extent to which they compete for resources. The niche overlap for a species is the fraction of the area under its preference curve that is also under the other species' curve. The niches displayed in the figure are given by

$$n_1(x) = (x - 1)(3 - x) \qquad 1 \leqslant x \leqslant 3$$

$$n_2(x) = (x - 2)(4 - x) \qquad 2 \leqslant x \leqslant 4$$

Use the Evaluation Theorem to calculate the niche overlap for species 1.

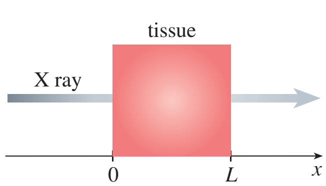


59. Medical imaging devices like CT scans work by passing an X-ray beam through part of the body and measuring how much the intensity of the beam attenuates (in other words, is reduced). The amount of attenuation depends on the density and composition of the tissue.

(a) If $A(x)$ is the attenuation rate at position x in the tissue and L is the thickness of the tissue through which the beam passes, what is the total attenuation of the beam during this procedure?

(b) Suppose the maximum and minimum attenuation rates in the tissue are β and α, respectively. Find upper and lower bounds for the total attenuation of the beam as it passes through the tissue. Justify your answer using the properties of integrals in Section 5.2.

Source: Adapted from T. Feeman, *The Mathematics of Medical Imaging* (New York: Springer Science + Business Media, 2011).



60. Incidence and prevalence The incidence $i(t)$ of an infectious disease at time t is the rate at which new infections are occurring at that time. The prevalence $P(t)$ at time t is the total number of infected individuals at that time. Let's suppose that $P(0) = 0$.

(a) Express the total number of new infections between times $t = 0$ and $t = a$ as a definite integral.

(b) Suppose that all individuals either die or recover from infection, and that D is the total number that have done so between times $t = 0$ and $t = a$. Express D in terms of $P(a)$ and your result from part (a).

(c) Let $d(t)$ be the rate at which people are dying or recovering from infection at time t. What is the relationship between D and $d(t)$?

61. Photosynthesis The *rate of primary production* refers to the rate of conversion of inorganic carbon to organic carbon via photosynthesis. It is measured as a mass of carbon fixed per unit biomass, per unit time. The rate of primary production depends on light intensity, measured as the flux of photons (that is, number of photons per unit area per unit time). One model for this relationship is

$$P(I) = P_{\max}(1 - e^{-aI})$$

where P is the rate of primary production as a function of light intensity I. Suppose that light intensity changes with time according to the equation $I(t) = kt$, where k is a constant.

(a) What is the rate of primary production as a function of time?

(b) What is the total amount of primary production over the first five units of time?

(c) What is the total amount of primary production over the first t units of time?

(d) What is the rate of change of total primary production at time t?

Source: Adapted from A. Jassby et al., "Mathematical Formulation of the Relationship between Photosynthesis and Light for Phytoplankton," *Limnology and Oceanography* 21 (1976): 540–7.

62. Photosynthesis Much of the earth's photosynthesis occurs in the oceans. The rate of primary production (as discussed in Exercise 61) depends on light intensity, measured as the flux of photons (that is, number of photons per unit area per unit time). For monochromatic light, intensity decreases with water depth according to Beer's Law, which states that $I(x) = e^{-kx}$, where x is water depth. A simple model for the relationship between rate of photosynthesis and light intensity is $P(I) = aI$, where a is a constant and P is measured as a mass of carbon fixed per volume of water, per unit time.

(a) What is the rate of photosynthesis as a function of water depth?

(b) What is the total rate of photosynthesis of a water column that is one unit in surface area and four units deep?

(c) What is the total rate of photosynthesis of a water column that is one unit in surface area and x units deep?

(d) What is the rate of change of the total photosynthesis with respect to the depth x?

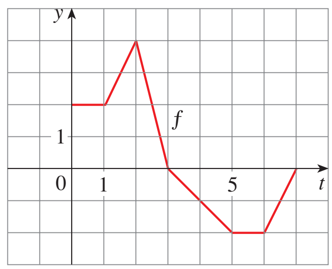
Source: Adapted from A. Jassby et al., "Mathematical Formulation of the Relationship between Photosynthesis and Light for Phytoplankton," *Limnology and Oceanography* 21 (1976): 540–7.

63. (a) Show that $1 \leqslant \sqrt{1 + x^3} \leqslant 1 + x^3$ for $x \geqslant 0$.

(b) Show that $1 \leqslant \int_0^1 \sqrt{1 + x^3}\, dx \leqslant 1.25$.

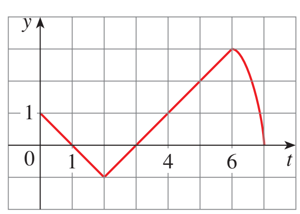
64. (a) Show that $\cos(x^2) \geqslant \cos x$ for $0 \leqslant x \leqslant 1$.
(b) Deduce that $\int_0^{\pi/6} \cos(x^2)\,dx \geqslant \frac{1}{2}$.

65. Let $g(x) = \int_0^x f(t)\,dt$, where f is the function whose graph is shown.



(a) Evaluate $g(0)$, $g(1)$, $g(2)$, $g(3)$, and $g(6)$.
(b) On what interval is g increasing?
(c) Where does g have a maximum value?
(d) Sketch a rough graph of g.

66. Let $g(x) = \int_0^x f(t)\,dt$, where f is the function whose graph is shown.
(a) Evaluate $g(x)$ for $x = 0, 1, 2, 3, 4, 5$, and 6.
(b) Estimate $g(7)$.
(c) Where does g have a maximum value? Where does it have a minimum value?
(d) Sketch a rough graph of g.



67–68 Sketch the area represented by $g(x)$. Then find $g'(x)$ in two ways: (a) by using Part 1 of the Fundamental Theorem and (b) by evaluating the integral using Part 2 and then differentiating.

67. $g(x) = \int_0^x (1 + t^2)\,dt$ **68.** $g(x) = \int_0^x (1 + \sqrt{t})\,dt$

69–78 Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of the function.

69. $g(x) = \int_1^x \dfrac{1}{t^3 + 1}\,dt$ **70.** $g(x) = \int_3^x e^{t^2 - t}\,dt$

71. $g(y) = \int_2^y t^2 \sin t\,dt$ **72.** $g(r) = \int_0^r \sqrt{x^2 + 4}\,dx$

73. $F(x) = \int_x^0 \sqrt{1 + \sec t}\,dt$

$\left[\textit{Hint}: \int_x^0 \sqrt{1 + \sec t}\,dt = -\int_0^x \sqrt{1 + \sec t}\,dt \right]$

74. $G(x) = \int_x^1 \cos\sqrt{t}\,dt$

75. $h(x) = \int_2^{1/x} \arctan t\,dt$ **76.** $h(x) = \int_0^{x^2} \sqrt{1 + r^3}\,dr$

77. $y = \int_0^{\tan x} \sqrt{t + \sqrt{t}}\,dt$ **78.** $y = \int_{e^x}^0 \sin^3 t\,dt$

79. If $f(1) = 12$, f' is continuous, and $\int_1^4 f'(x)\,dx = 17$, what is the value of $f(4)$?

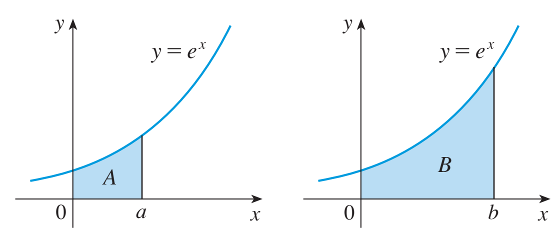
80. The *error function*

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt$$

is used in probability, statistics, and engineering.
(a) Show that $\int_a^b e^{-t^2}\,dt = \frac{1}{2}\sqrt{\pi}\,[\operatorname{erf}(b) - \operatorname{erf}(a)]$.
(b) Show that the function $y = e^{x^2}\operatorname{erf}(x)$ satisfies the differential equation $y' = 2xy + 2/\sqrt{\pi}$.

81. Suppose h is a function such that $h(1) = -2$, $h'(1) = 2$, $h''(1) = 3$, $h(2) = 6$, $h'(2) = 5$, $h''(2) = 13$, and h'' is continuous everywhere. Evaluate $\int_1^2 h''(u)\,du$.

82. The area labeled B is three times the area labeled A. Express b in terms of a.



83–84 Evaluate the limit by first recognizing the sum as a Riemann sum for a function defined on [0, 1].

83. $\displaystyle\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^3}{n^4}$

84. $\displaystyle\lim_{n\to\infty} \frac{1}{n}\left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \sqrt{\frac{3}{n}} + \cdots + \sqrt{\frac{n}{n}}\right)$

85. Find a function f and a number a such that

$$6 + \int_a^x \frac{f(t)}{t^2}\,dt = 2\sqrt{x} \qquad \text{for all } x > 0$$

PROJECT The Outbreak Size of an Infectious Disease

BB

In Sections 7.6 and 10.4 we will analyze the Kermack-McKendrick model for infectious disease dynamics. If $S(t)$ is the number of people susceptible to infection at time t and $I(t)$ is the number of people already infected at this time, then a simplified version of the model specifies the derivatives of S and I with respect to time as

(1) $$\frac{dS}{dt} = -\beta SI$$

(2) $$\frac{dI}{dt} = \beta SI - \mu I$$

where β and μ are positive constants and the numbers of susceptible and infected people at time $t = 0$ are $S(0)$ and $I(0)$, respectively.

1. Use Equation 1 to express the quantity $-\beta \int_0^T S(t)I(t)\,dt$ in terms of the function $S(t)$.
2. Use Equation 1 to express the quantity $-\beta \int_0^T I(t)\,dt$ in terms of the function $S(t)$.
3. Use Equation 2 to express the quantity $\beta \int_0^T S(t)I(t)\,dt - \mu \int_0^T I(t)\,dt$ in terms of the function $I(t)$.
4. Use the results from Problems 1–3 to obtain a single equation that $S(0)$, $S(T)$, $I(T)$, and $I(0)$ must satisfy and that does not involve integrals.
5. It can be shown that $\lim_{T\to\infty} I(T) = 0$ and that $\lim_{T\to\infty} S(T) = S_\infty$, where S_∞ is a positive constant. What equation do you get if you let $T \to \infty$ in your answer to Problem 4?
6. Suppose that the number of people initially infected, $I(0)$, is negligibly small and define $q = \beta S(0)/\mu$ and $A = 1 - S_\infty/S(0)$. Here q is a measure of the transmissibility of the infection and A is the fraction of the original susceptible population $S(0)$ that ultimately gets infected (that is, the outbreak size). Show that your answer to Problem 5 can be written as

   (3) $$e^{-qA} = 1 - A$$

   Equation 3 is a special case of an equation seen previously in many exercises and examples. For instance, see Example 3.5.13, Exercises 3.5.81, 3.8.32, 3.Review.92, and Example 9.4.6.

5.4 The Substitution Rule

Substitution in Indefinite Integrals

Because of the Fundamental Theorem, it's important to be able to find antiderivatives. But our antidifferentiation formulas don't tell us how to evaluate integrals such as

(1) $$\int 2x\sqrt{1 + x^2}\,dx$$

Our strategy is to simplify the integral by introducing a new variable. Suppose that we let u be the quantity under the root sign in (1): $u = 1 + x^2$. Then $du/dx = 2x$. Up until now we have not interpreted du/dx as a ratio of two quantities du and dx, but in this context it is useful to do so. Here we regard the symbols du and dx as separate entities called *differentials* and we write the equation $du/dx = 2x$ as

$$du = 2x\,dx$$

Then formally, without justifying our calculation, we could write

(2)
$$\begin{aligned}\int 2x\sqrt{1+x^2}\,dx &= \int \sqrt{1+x^2}\,2x\,dx = \int \sqrt{u}\,du \\ &= \tfrac{2}{3}u^{3/2} + C = \tfrac{2}{3}(1+x^2)^{3/2} + C\end{aligned}$$

But now we can check that we have the correct answer by using the Chain Rule to differentiate the final function of Equation 2:

$$\frac{d}{dx}\left[\tfrac{2}{3}(1+x^2)^{3/2} + C\right] = \tfrac{2}{3}\cdot\tfrac{3}{2}(1+x^2)^{1/2}\cdot 2x = 2x\sqrt{1+x^2}$$

In general, this method works whenever we have an integral that we can write in the form $\int f(g(x))\,g'(x)\,dx$. Observe that if $F' = f$, then

(3)
$$\int F'(g(x))\,g'(x)\,dx = F(g(x)) + C$$

because, by the Chain Rule,

$$\frac{d}{dx}[F(g(x))] = F'(g(x))\,g'(x)$$

If we make the "change of variable" or "substitution" $u = g(x)$, then from Equation 3 we have

$$\int F'(g(x))\,g'(x)\,dx = F(g(x)) + C = F(u) + C = \int F'(u)\,du$$

or, writing $F' = f$, we get

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du$$

Thus we have proved the following rule.

(4) The Substitution Rule If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I, then

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du$$

Notice that the Substitution Rule for integration was proved using the Chain Rule for differentiation. Notice also that if $u = g(x)$, then $du/dx = g'(x)$, which we write in terms of **differentials**: $du = g'(x)\,dx$. This is probably the best way to remember the Substitution Rule.

EXAMPLE 1 | Find $\int x^3\cos(x^4 + 2)\,dx$.

SOLUTION We make the substitution $u = x^4 + 2$ because its differential is $du = 4x^3\,dx$, which, apart from the constant factor 4, occurs in the integral. Thus, using $x^3\,dx = \frac{1}{4}\,du$ and the Substitution Rule, we have

$$\begin{aligned}\int x^3\cos(x^4+2)\,dx &= \int \cos u\cdot\tfrac{1}{4}\,du = \tfrac{1}{4}\int \cos u\,du\\ &= \tfrac{1}{4}\sin u + C\\ &= \tfrac{1}{4}\sin(x^4+2)+C\end{aligned}$$

Check the answer by differentiating it.

Notice that at the final stage we had to return to the original variable x. ■

The idea behind the Substitution Rule is to replace a relatively complicated integral by a simpler integral. This is accomplished by changing from the original variable x to a new variable u that is a function of x. Thus in Example 1 we replaced the integral $\int x^3\cos(x^4+2)\,dx$ by the simpler integral $\frac{1}{4}\int \cos u\,du$.

The main challenge in using the Substitution Rule is to think of an appropriate substitution. You should try to choose u to be some function in the integrand whose differential also occurs (except for a constant factor). This was the case in Example 1. If that is not possible, try choosing u to be some complicated part of the integrand (perhaps the inner function in a composite function). Finding the right substitution is a bit of an art. It's not unusual to guess wrong; if your first guess doesn't work, try another substitution.

EXAMPLE 2 | Evaluate $\int \sqrt{2x+1}\,dx$.

SOLUTION 1 Let $u = 2x+1$. Then $du = 2\,dx$, so $dx = \frac{1}{2}\,du$. Thus the Substitution Rule gives

$$\begin{aligned}\int \sqrt{2x+1}\,dx &= \int \sqrt{u}\cdot\tfrac{1}{2}\,du = \tfrac{1}{2}\int u^{1/2}\,du\\ &= \frac{1}{2}\cdot\frac{u^{3/2}}{3/2} + C = \tfrac{1}{3}u^{3/2}+C\\ &= \tfrac{1}{3}(2x+1)^{3/2}+C\end{aligned}$$

SOLUTION 2 Another possible substitution is $u = \sqrt{2x+1}$. Then

$$du = \frac{dx}{\sqrt{2x+1}} \qquad \text{so} \qquad dx = \sqrt{2x+1}\,du = u\,du$$

(Or observe that $u^2 = 2x+1$, so $2u\,du = 2\,dx$.) Therefore

$$\begin{aligned}\int \sqrt{2x+1}\,dx &= \int u\cdot u\,du = \int u^2\,du\\ &= \frac{u^3}{3} + C = \tfrac{1}{3}(2x+1)^{3/2}+C\end{aligned}$$ ■

EXAMPLE 3 | Find $\displaystyle\int \frac{x}{\sqrt{1-4x^2}}\,dx$.

SOLUTION Let $u = 1-4x^2$. Then $du = -8x\,dx$, so $x\,dx = -\frac{1}{8}\,du$ and

$$\begin{aligned}\int \frac{x}{\sqrt{1-4x^2}}\,dx &= -\tfrac{1}{8}\int \frac{1}{\sqrt{u}}\,du = -\tfrac{1}{8}\int u^{-1/2}\,du\\ &= -\tfrac{1}{8}\left(2\sqrt{u}\right)+C = -\tfrac{1}{4}\sqrt{1-4x^2}+C\end{aligned}$$ ■

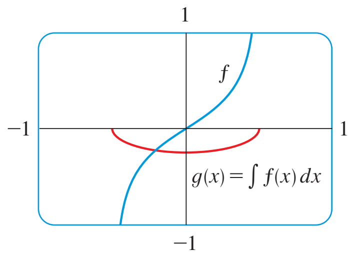


FIGURE 1

$f(x) = \dfrac{x}{\sqrt{1-4x^2}}$

$g(x) = \int f(x)\,dx = -\frac{1}{4}\sqrt{1-4x^2}$

The answer to Example 3 could be checked by differentiation, but instead let's check it with a graph. In Figure 1 we have used a computer to graph both the integrand $f(x) = x/\sqrt{1-4x^2}$ and its indefinite integral $g(x) = -\frac{1}{4}\sqrt{1-4x^2}$ (we take the case $C = 0$). Notice that $g(x)$ decreases when $f(x)$ is negative, increases when $f(x)$ is positive, and has its minimum value when $f(x) = 0$. So it seems reasonable, from the graphical evidence, that g is an antiderivative of f.

EXAMPLE 4 | Calculate $\int e^{5x}\,dx$.

SOLUTION If we let $u = 5x$, then $du = 5\,dx$, so $dx = \frac{1}{5}\,du$. Therefore

$$\int e^{5x}\,dx = \tfrac{1}{5}\int e^u\,du = \tfrac{1}{5}e^u + C = \tfrac{1}{5}e^{5x} + C$$ ■

NOTE With some experience, you might be able to evaluate integrals like those in Examples 1–4 without going to the trouble of making an explicit substitution. By recognizing the pattern in Equation 3, where the integrand on the left side is the product of the derivative of an outer function and the derivative of the inner function, we could work Example 1 as follows:

$$\int x^3\cos(x^4+2)\,dx = \int \cos(x^4+2)\cdot x^3\,dx = \tfrac{1}{4}\int \cos(x^4+2)\cdot(4x^3)\,dx$$

$$= \tfrac{1}{4}\int \cos(x^4+2)\cdot\frac{d}{dx}(x^4+2)\,dx = \tfrac{1}{4}\sin(x^4+2) + C$$

Similarly, the solution to Example 4 could be written like this:

$$\int e^{5x}\,dx = \tfrac{1}{5}\int 5e^{5x}\,dx = \tfrac{1}{5}\int \frac{d}{dx}(e^{5x})\,dx = \tfrac{1}{5}e^{5x} + C$$

The following example, however, is more complicated and so an explicit substitution is advisable.

EXAMPLE 5 | Calculate $\int \tan x\,dx$.

SOLUTION First we write tangent in terms of sine and cosine:

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$$

This suggests that we should substitute $u = \cos x$, since then $du = -\sin x\,dx$ and so $\sin x\,dx = -du$:

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{1}{u}\,du$$

$$= -\ln|u| + C = -\ln|\cos x| + C$$ ■

Since $-\ln|\cos x| = \ln(|\cos x|^{-1}) = \ln(1/|\cos x|) = \ln|\sec x|$, the result of Example 5 can also be written as

$$\int \tan x\,dx = \ln|\sec x| + C$$

Substitution in Definite Integrals

When evaluating a *definite* integral by substitution, two methods are possible. One method is to evaluate the indefinite integral first and then use the Evaluation Theorem. For instance, using the result of Example 2, we have

$$\begin{aligned}\int_0^4 \sqrt{2x+1}\,dx &= \int \sqrt{2x+1}\,dx\Big]_0^4 \\ &= \tfrac{1}{3}(2x+1)^{3/2}\Big]_0^4 = \tfrac{1}{3}(9)^{3/2} - \tfrac{1}{3}(1)^{3/2} \\ &= \tfrac{1}{3}(27-1) = \tfrac{26}{3}\end{aligned}$$

Another method, which is usually preferable, is to change the limits of integration when the variable is changed.

This rule says that when using a substitution in a definite integral, we must put everything in terms of the new variable u, not only x and dx but also the limits of integration. The new limits of integration are the values of u that correspond to $x = a$ and $x = b$.

(5) The Substitution Rule for Definite Integrals If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

PROOF Let F be an antiderivative of f. Then, by (3), $F(g(x))$ is an antiderivative of $f(g(x))\,g'(x)$, so by the Evaluation Theorem, we have

$$\int_a^b f(g(x))\,g'(x)\,dx = F(g(x))\Big]_a^b = F(g(b)) - F(g(a))$$

But, applying the Evaluation Theorem a second time, we also have

$$\int_{g(a)}^{g(b)} f(u)\,du = F(u)\Big]_{g(a)}^{g(b)} = F(g(b)) - F(g(a))$$ ■

EXAMPLE 6 | Evaluate $\int_0^4 \sqrt{2x+1}\,dx$ using (5).

SOLUTION Using the substitution from Solution 1 of Example 2, we have $u = 2x + 1$ and $dx = \frac{1}{2}\,du$. To find the new limits of integration we note that

when $x = 0$, $u = 2(0) + 1 = 1$ and when $x = 4$, $u = 2(4) + 1 = 9$

Therefore

$$\begin{aligned}\int_0^4 \sqrt{2x+1}\,dx &= \int_1^9 \tfrac{1}{2}\sqrt{u}\,du = \tfrac{1}{2}\cdot\tfrac{2}{3}u^{3/2}\Big]_1^9 \\ &= \tfrac{1}{3}(9^{3/2} - 1^{3/2}) = \tfrac{26}{3}\end{aligned}$$

Observe that when using (5) we do *not* return to the variable x after integrating. We simply evaluate the expression in u between the appropriate values of u. ■

The integral given in Example 7 is an abbreviation for

$$\int_1^2 \frac{1}{(3-5x)^2}\,dx$$

EXAMPLE 7 | Evaluate $\displaystyle\int_1^2 \frac{dx}{(3-5x)^2}$.

SOLUTION Let $u = 3 - 5x$. Then $du = -5\,dx$, so $dx = -\frac{1}{5}\,du$. When $x = 1$,

$u = -2$ and when $x = 2$, $u = -7$. Thus

$$\int_1^2 \frac{dx}{(3-5x)^2} = -\frac{1}{5}\int_{-2}^{-7}\frac{du}{u^2}$$

$$= -\frac{1}{5}\left[-\frac{1}{u}\right]_{-2}^{-7} = \frac{1}{5u}\Bigg]_{-2}^{-7}$$

$$= \frac{1}{5}\left(-\frac{1}{7}+\frac{1}{2}\right) = \frac{1}{14}$$ ■

EXAMPLE 8 | Metabolism A model for the basal metabolism rate, in kcal/h, of a young man is

$$R(t) = 85 - 0.18\cos(\pi t/12)$$

where t is the time in hours measured from 5:00 AM. What is the total basal metabolism of this man, $\int_0^{24} R(t)\,dt$, over a 24-hour time period?

SOLUTION We make the substitution $u = \pi t/12$. Then $du = (\pi/12)\,dt$ and $u = 2\pi$ when $t = 24$. So

$$\int_0^{24} R(t)\,dt = \int_0^{24}\left[85 - 0.18\cos\left(\frac{\pi t}{12}\right)\right]dt = \int_0^{2\pi}(8.5 - 0.18\cos u)\cdot\frac{12}{\pi}\,du$$

$$= \frac{12}{\pi}\Big[85u - 0.18\sin u\Big]_0^{2\pi} = \frac{12}{\pi}(85\cdot 2\pi - 0) = 2040$$

The total metabolism for the 24-hour period was 2040 kcal. ■

Symmetry

The next theorem uses the Substitution Rule for Definite Integrals (5) to simplify the calculation of integrals of functions that possess symmetry properties.

(6) Integrals of Symmetric Functions Suppose f is continuous on $[-a, a]$.

(a) If f is even $[f(-x) = f(x)]$, then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.

(b) If f is odd $[f(-x) = -f(x)]$, then $\int_{-a}^{a} f(x)\,dx = 0$.

PROOF We split the integral in two:

$$\textbf{(7)} \qquad \int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_0^a f(x)\,dx = -\int_0^{-a} f(x)\,dx + \int_0^a f(x)\,dx$$

In the first integral on the far right side we make the substitution $u = -x$. Then $du = -dx$ and when $x = -a$, $u = a$. Therefore

$$-\int_0^{-a} f(x)\,dx = -\int_0^a f(-u)\,(-du) = \int_0^a f(-u)\,du$$

and so Equation 7 becomes

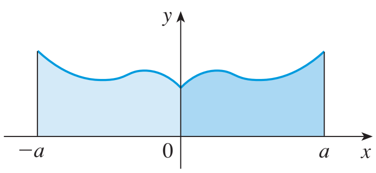
$$\textbf{(8)} \qquad \int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(-u)\,du + \int_{0}^{a} f(x)\,dx$$

(a) If f is even, then $f(-u) = f(u)$ so Equation 8 gives

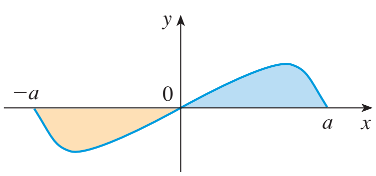
$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(u)\,du + \int_{0}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

(b) If f is odd, then $f(-u) = -f(u)$ and so Equation 8 gives

$$\int_{-a}^{a} f(x)\,dx = -\int_{0}^{a} f(u)\,du + \int_{0}^{a} f(x)\,dx = 0$$



(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

FIGURE 2

Theorem 6 is illustrated by Figure 2. For the case where f is positive and even, part (a) says that the area under $y = f(x)$ from $-a$ to a is twice the area from 0 to a because of symmetry. Recall that an integral $\int_a^b f(x)\,dx$ can be expressed as the area above the x-axis and below $y = f(x)$ minus the area below the axis and above the curve. Thus part (b) says the integral is 0 because the areas cancel.

EXAMPLE 9 | Since $f(x) = x^6 + 1$ satisfies $f(-x) = f(x)$, it is even and so

$$\int_{-2}^{2} (x^6 + 1)\,dx = 2\int_{0}^{2} (x^6 + 1)\,dx$$

$$= 2\Big[\tfrac{1}{7}x^7 + x\Big]_0^2 = 2\left(\tfrac{128}{7} + 2\right) = \tfrac{284}{7}$$

EXAMPLE 10 | Since $f(x) = (\tan x)/(1 + x^2 + x^4)$ satisfies $f(-x) = -f(x)$, it is odd and so

$$\int_{-1}^{1} \frac{\tan x}{1 + x^2 + x^4}\,dx = 0$$

EXERCISES 5.4

1–6 Evaluate the integral by making the given substitution.

1. $\int e^{-x}\,dx, \quad u = -x$

2. $\int x^3(2 + x^4)^5\,dx, \quad u = 2 + x^4$

3. $\int x^2\sqrt{x^3 + 1}\,dx, \quad u = x^3 + 1$

4. $\int \dfrac{dt}{(1 - 6t)^4}, \quad u = 1 - 6t$

5. $\int \cos^3\theta\,\sin\theta\,d\theta, \quad u = \cos\theta$

6. $\int \dfrac{\sec^2(1/x)}{x^2}\,dx, \quad u = 1/x$

7–36 Evaluate the indefinite integral.

7. $\int x\sin(x^2)\,dx$

8. $\int x^2(x^3 + 5)^9\,dx$

9. $\int (3x - 2)^{20}\,dx$

10. $\int (3t + 2)^{2.4}\,dt$

11. $\int \sin \pi t\,dt$

12. $\int e^x\cos(e^x)\,dx$

13. $\int \dfrac{(\ln x)^2}{x}\,dx$

14. $\int \dfrac{x}{(x^2 + 1)^2}\,dx$

15. $\int \dfrac{dx}{5 - 3x}$

16. $\int \dfrac{\sin\sqrt{x}}{\sqrt{x}}\,dx$

17. $\int \dfrac{a + bx^2}{\sqrt{3ax + bx^3}}\,dx$

18. $\int \dfrac{z^2}{z^3 + 1}\,dz$

19. $\int e^x\sqrt{1 + e^x}\,dx$

20. $\int \sec 2\theta\,\tan 2\theta\,d\theta$

21. $\int \dfrac{\cos x}{\sin^2 x}\,dx$

22. $\int \dfrac{\tan^{-1}x}{1 + x^2}\,dx$

23. $\int (x^2 + 1)(x^3 + 3x)^4\,dx$

24. $\int \dfrac{\sin(\ln x)}{x}\,dx$

25. $\int \sqrt{\cot x}\, \csc^2 x\, dx$

26. $\int \dfrac{\cos(\pi/x)}{x^2}\, dx$

27. $\int e^{2r} \sin(e^{2r})\, dr$

28. $\int \dfrac{dt}{\cos^2 t \sqrt{1 + \tan t}}$

29. $\int \sec^3 x \tan x\, dx$

30. $\int x^2 \sqrt{2 + x}\, dx$

31. $\int x(2x + 5)^8\, dx$

32. $\int \dfrac{e^x}{e^x + 1}\, dx$

33. $\int \dfrac{\sin 2x}{1 + \cos^2 x}\, dx$

34. $\int \dfrac{\sin x}{1 + \cos^2 x}\, dx$

35. $\int \dfrac{1 + x}{1 + x^2}\, dx$

36. $\int \dfrac{x}{1 + x^4}\, dx$

37–51 Evaluate the definite integral.

37. $\int_0^1 \cos(\pi t/2)\, dt$

38. $\int_0^1 (3t - 1)^{50}\, dt$

39. $\int_0^1 \sqrt[3]{1 + 7x}\, dx$

40. $\int_0^{\sqrt{\pi}} x \cos(x^2)\, dx$

41. $\int_0^1 x^2(1 + 2x^3)^5\, dx$

42. $\int_{1/6}^{1/2} \csc \pi t \cot \pi t\, dt$

43. $\int_1^4 \dfrac{e^{\sqrt{x}}}{\sqrt{x}}\, dx$

44. $\int_0^{\pi/2} \cos x \sin(\sin x)\, dx$

45. $\int_{-\pi/4}^{\pi/4} (x^3 + x^4 \tan x)\, dx$

46. $\int_{-\pi/2}^{\pi/2} \dfrac{x^2 \sin x}{1 + x^6}\, dx$

47. $\int_1^2 x\sqrt{x - 1}\, dx$

48. $\int_0^a x\sqrt{a^2 - x^2}\, dx$

49. $\int_0^1 \dfrac{e^z + 1}{e^z + z}\, dz$

50. $\int_0^{T/2} \sin(2\pi t/T - \alpha)\, dt$

51. $\int_0^1 \dfrac{dx}{(1 + \sqrt{x})^4}$

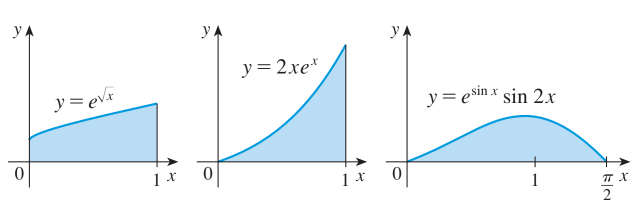
52. Verify that $f(x) = \sin \sqrt[3]{x}$ is an odd function and use that fact to show that

$$0 \leqslant \int_{-2}^{3} \sin \sqrt[3]{x}\, dx \leqslant 1$$

53. Evaluate $\int_{-2}^{2} (x + 3)\sqrt{4 - x^2}\, dx$ by writing it as a sum of two integrals and interpreting one of those integrals in terms of an area.

54. Evaluate $\int_0^1 x\sqrt{1 - x^4}\, dx$ by making a substitution and interpreting the resulting integral in terms of an area.

55. Which of the following areas are equal? Why?



56. A **bacteria population** starts with 400 bacteria and grows at a rate of $r(t) = (450.268)e^{1.12567t}$ bacteria per hour. How many bacteria will there be after three hours?

57. An oil storage tank ruptures at time $t = 0$ and oil leaks from the tank at a rate of $r(t) = 100e^{-0.01t}$ liters per minute. How much oil leaks out during the first hour?

58. Fish biomass The rate of growth of a fish population was modeled by the equation

$$G(t) = \frac{60{,}000e^{-0.6t}}{(1 + 5e^{-0.6t})^2}$$

where t is measured in years since 2000 and G in kilograms per year. If the biomass was 25,000 kg in the year 2000, what is the predicted biomass for the year 2020?

59. Breathing is cyclic and a full respiratory cycle from the beginning of inhalation to the end of exhalation takes about 5 s. The maximum rate of air flow into the lungs is about 0.5 L/s. This explains, in part, why the function $f(t) = \frac{1}{2}\sin(2\pi t/5)$ has often been used to model the rate of air flow into the lungs. Use this model to find the volume of inhaled air in the lungs at time t.

60. Dialysis treatment removes urea and other waste products from a patient's blood by diverting some of the bloodflow externally through a machine called a dialyzer. The rate at which urea is removed from the blood (in mg/min) is often well described by the equation

$$c(t) = \frac{K}{V} c_0 e^{-Kt/V}$$

where K is the rate of flow of blood through the dialyzer (in mL/min), V is the volume of the patient's blood (in mL), and c_0 is the amount of urea in the blood (in mg) at time $t = 0$. Evaluate the integral $\int_0^{30} c(t)\, dt$ and interpret it.

61. Gompertz tumor growth In Chapter 7 we will explore a model for tumor growth in which the growth rate is given by

$$g(t) = 2^{1-e^{-t}} e^{-t} \ln 2 \text{ mm}^3\text{/month}$$

By how much is the volume of the tumor predicted to increase over the first year?

62. Photosynthesis The *rate of primary production* refers to the rate of conversion of inorganic carbon to organic carbon via photosynthesis. It is measured as a mass of carbon fixed per unit biomass, per unit time. A common model for this relationship is

$$P(I) = \frac{aI}{\sqrt{1 + bI^2}}$$

where P is the rate of primary production as a function of light intensity I. Suppose the light intensity changes with time according to the equation $I(t) = kt$, where k is a constant.

(a) What is the rate of primary production as a function of time?

(b) What is the total amount of primary production over the first five units of time?

Source: Adapted from A. Jassby et al., "Mathematical Formulation of the Relationship between Photosynthesis and Light for Phytoplankton," *Limnology and Oceanography* 21 (1976): 540–7.

63. Growing degree days The rate of development of many plant species depends on the temperature of the environment in such a way that maturity is reached only after a certain number of "degree-days." Suppose that temperature T as a function of time t is given by

$$T(t) = 15\left(1 + \sin\frac{2\pi t}{60}\right) \qquad 0 \leqslant t \leqslant 60$$

where t is measured in days. If maturity is reached on day $t = 20$, what is the number of degree-days required? [*Hint:* What are the units for $\int_0^{20} T(t)\,dt$?]

64. If f is continuous and $\int_0^9 f(x)\,dx = 4$, find $\int_0^3 xf(x^2)\,dx$.

65. If f is continuous and $\int_0^4 f(x)\,dx = 10$, find $\int_0^2 f(2x)\,dx$.

66. If f is continuous on $\mathbb{R}$, prove that

$$\int_a^b f(x + c)\,dx = \int_{a+c}^{b+c} f(x)\,dx$$

For the case where $f(x) \geqslant 0$, draw a diagram to interpret this equation geometrically as an equality of areas.

67. If a and b are positive numbers, show that

$$\int_0^1 x^a(1 - x)^b\,dx = \int_0^1 x^b(1 - x)^a\,dx$$

5.5 Integration by Parts

Every differentiation rule has a corresponding integration rule. For instance, the Substitution Rule for integration corresponds to the Chain Rule for differentiation. The rule that corresponds to the Product Rule for differentiation is called the rule for *integration by parts*.

■ Indefinite Integrals

The Product Rule states that if f and g are differentiable functions, then

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

In the notation for indefinite integrals this equation becomes

$$\int [f(x)g'(x) + g(x)f'(x)]\,dx = f(x)g(x)$$

or

$$\int f(x)g'(x)\,dx + \int g(x)f'(x)\,dx = f(x)g(x)$$

We can rearrange this equation as

(1)

$$\boxed{\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx}$$

Formula 1 is called the **formula for integration by parts**. It is perhaps easier to remember in the following notation. Let $u = f(x)$ and $v = g(x)$. Then the differentials are $du = f'(x)\,dx$ and $dv = g'(x)\,dx$, so, by the Substitution Rule, the formula for integration by parts becomes

(2)

$$\boxed{\int u\,dv = uv - \int v\,du}$$

EXAMPLE 1 | Find $\int x \sin x\,dx$.

SOLUTION USING FORMULA 1 Suppose we choose $f(x) = x$ and $g'(x) = \sin x$. Then $f'(x) = 1$ and $g(x) = -\cos x$. (For g we can choose *any* antiderivative of g'.) Thus, using Formula 1, we have

$$\begin{aligned}\int x \sin x\,dx &= f(x)g(x) - \int g(x)f'(x)\,dx \\ &= x(-\cos x) - \int (-\cos x)\,dx \\ &= -x\cos x + \int \cos x\,dx \\ &= -x\cos x + \sin x + C\end{aligned}$$

It's wise to check the answer by differentiating it. If we do so, we get $x \sin x$, as expected.

It is helpful to use the pattern:

$$\begin{array}{ll} u = \square & dv = \square \\ du = \square & v = \square \end{array}$$

SOLUTION USING FORMULA 2 Let

$$u = x \qquad dv = \sin x\,dx$$

Then $$du = dx \qquad v = -\cos x$$

and so

$$\begin{aligned}\int x \sin x\,dx = \int \overbrace{x}^{u}\ \overbrace{\sin x\,dx}^{dv} &= \overbrace{x}^{u}\ \overbrace{(-\cos x)}^{v} - \int \overbrace{(-\cos x)}^{v}\ \overbrace{dx}^{du} \\ &= -x\cos x + \int \cos x\,dx \\ &= -x\cos x + \sin x + C\end{aligned}$$

■

NOTE Our aim in using integration by parts is to obtain a simpler integral than the one we started with. Thus in Example 1 we started with $\int x \sin x\,dx$ and expressed it in terms of the simpler integral $\int \cos x\,dx$. If we had instead chosen $u = \sin x$ and $dv = x\,dx$, then $du = \cos x\,dx$ and $v = x^2/2$, so integration by parts gives

$$\int x \sin x\,dx = (\sin x)\frac{x^2}{2} - \frac{1}{2}\int x^2 \cos x\,dx$$

Although this is true, $\int x^2 \cos x\,dx$ is a more difficult integral than the one we started with. In general, when deciding on a choice for u and dv, we usually try to choose $u = f(x)$ to be a function that becomes simpler when differentiated (or at least not more complicated) as long as $dv = g'(x)\,dx$ can be readily integrated to give v.

EXAMPLE 2 | Evaluate $\int \ln x\,dx$.

SOLUTION Here we don't have much choice for u and dv. Let

$$u = \ln x \qquad dv = dx$$

Then $$du = \frac{1}{x}\,dx \qquad v = x$$

Integrating by parts, we get

$$\int \ln x\,dx = x\ln x - \int x\,\frac{dx}{x}$$
$$= x\ln x - \int dx$$
$$= x\ln x - x + C$$

It's customary to write $\int 1\,dx$ as $\int dx$.

Check the answer by differentiating it.

Integration by parts is effective in this example because the derivative of the function $f(x) = \ln x$ is simpler than f. ■

EXAMPLE 3 | Find $\int t^2 e^t\,dt$.

SOLUTION Notice that t^2 becomes simpler when differentiated (whereas e^t is unchanged when differentiated or integrated), so we choose

$$u = t^2 \qquad dv = e^t\,dt$$

Then $$du = 2t\,dt \qquad v = e^t$$

Integration by parts gives

$$\int t^2 e^t\,dt = t^2 e^t - 2\int te^t\,dt \tag{3}$$

The integral that we obtained, $\int te^t\,dt$, is simpler than the original integral but is still not obvious. Therefore we use integration by parts a second time, this time with $u = t$ and $dv = e^t\,dt$. Then $du = dt$, $v = e^t$, and

$$\int te^t\,dt = te^t - \int e^t\,dt$$
$$= te^t - e^t + C$$

Putting this in Equation 3, we get

$$\int t^2 e^t\,dt = t^2 e^t - 2\int te^t\,dt$$
$$= t^2 e^t - 2(te^t - e^t + C)$$
$$= t^2 e^t - 2te^t + 2e^t + C_1 \qquad \text{where } C_1 = -2C$$ ■

EXAMPLE 4 | Evaluate $\int e^x \sin x\,dx$.

An easier method, using complex numbers, is given in Exercise 50 in Appendix G.

SOLUTION Neither e^x nor $\sin x$ becomes simpler when differentiated, but we try choosing $u = e^x$ and $dv = \sin x\,dx$ anyway. Then $du = e^x\,dx$ and $v = -\cos x$, so integration by parts gives

$$\int e^x \sin x\,dx = -e^x\cos x + \int e^x \cos x\,dx \tag{4}$$

The integral that we have obtained, $\int e^x \cos x\,dx$, is no simpler than the original one, but at least it's no more difficult. Having had success in the preceding example integrating by parts twice, we persevere and integrate by parts again. This time we use $u = e^x$ and $dv = \cos x\,dx$. Then $du = e^x\,dx$, $v = \sin x$, and

$$\int e^x \cos x\,dx = e^x \sin x - \int e^x \sin x\,dx \tag{5}$$

Figure 1 illustrates Example 4 by showing the graphs of $f(x) = e^x \sin x$ and $F(x) = \frac{1}{2}e^x(\sin x - \cos x)$. As a visual check on our work, notice that $f(x) = 0$ when F has a maximum or minimum.

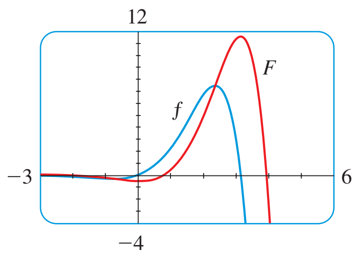


FIGURE 1

At first glance, it appears as if we have accomplished nothing because we have arrived at $\int e^x \sin x\,dx$, which is where we started. However, if we put the expression for $\int e^x \cos x\,dx$ from Equation 5 into Equation 4 we get

$$\int e^x \sin x\,dx = -e^x \cos x + e^x \sin x - \int e^x \sin x\,dx$$

This can be regarded as an equation to be solved for the unknown integral. Adding $\int e^x \sin x\,dx$ to both sides, we obtain

$$2\int e^x \sin x\,dx = -e^x \cos x + e^x \sin x$$

Dividing by 2 and adding the constant of integration, we get

$$\int e^x \sin x\,dx = \tfrac{1}{2}e^x(\sin x - \cos x) + C \qquad \blacksquare$$

■ Definite Integrals

If we combine the formula for integration by parts with the Evaluation Theorem, we can evaluate definite integrals by parts. Evaluating both sides of Formula 1 between a and b, assuming f' and g' are continuous, and using the Evaluation Theorem, we obtain

(6) $$\boxed{\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big]_a^b - \int_a^b g(x)f'(x)\,dx}$$

EXAMPLE 5 | Aspirin pharmacokinetics In Example 5.2.5 we used data from a paper[1] and modeled the average concentration of low-dose aspirin in the bloodstream of 10 volunteers by the function

$$C(t) = 32t^2e^{-4.2t}$$

where t is measured in hours and C is measured in μg/mL. There we used the Midpoint Rule to estimate $\int_0^2 C(t)\,dt$ and interpreted the integral in terms of the availability of the drug. Here we use integration by parts to evaluate $\int_0^2 C(t)\,dt$.

SOLUTION Notice that t^2 becomes simpler when differentiated (whereas $e^{-4.2t}$ doesn't). So we choose

$$u = 32t^2 \qquad dv = e^{-4.2t}\,dt$$

Then

$$du = 64t\,dt \qquad v = -\frac{1}{4.2}e^{-4.2t}$$

Formula 6 gives

(7) $$\int_0^2 32t^2e^{-4.2t}\,dt = -\frac{32}{4.2}t^2e^{-4.2t}\Big]_0^2 - \int_0^2\left(-\frac{64}{4.2}\right)te^{-4.2t}\,dt$$

$$= -\frac{32}{4.2}(4e^{-8.4}) + \frac{64}{4.2}\int_0^2 te^{-4.2t}\,dt$$

1. I. H. Benedek et al., "Variability in the Pharmacokinetics and Pharmacodynamics of Low Dose Aspirin in Healthy Male Volunteers," *Journal of Clinical Pharmacology* 35 (1995): 1181–86.

The integral that we obtained, $\int_0^2 te^{-4.2t}\,dt$, is simpler than the original integral but is still not obvious. Therefore we use integration by parts a second time, this time with $u = t$ and $dv = e^{-4.2t}\,dt$. Then

$$du = dt \qquad \text{and} \qquad v = -\frac{1}{4.2}e^{-4.2t}$$

so

$$\int_0^2 te^{-4.2t}\,dt = -\frac{t}{4.2}\,e^{-4.2t}\Big]_0^2 + \frac{1}{4.2}\int_0^2 e^{-4.2t}\,dt$$

$$= -\frac{2}{4.2}\,e^{-8.4} + \frac{1}{4.2}\left[\frac{e^{-4.2t}}{-4.2}\right]_0^2 = -\frac{2}{4.2}\,e^{-8.4} - \frac{e^{-8.4}-1}{(4.2)^2}$$

Putting this in Equation 7, we get

$$\int_0^2 C(t)\,dt = \frac{1}{4.2}\left[-128e^{-8.4} + 64\left(-\frac{2}{4.2}e^{-8.4} - \frac{e^{-8.4}-1}{(4.2)^2}\right)\right]$$

Comparing the answer in Example 5 with the estimate of 0.8568 that we got in Example 5.2.5, we see that the Midpoint Rule gave a reasonably accurate estimate.

Approximating this expression to four decimal places, we get

$$\int_0^2 C(t)\,dt \approx 0.8552\ (\mu\text{g/mL}) \cdot \text{h}$$ ■

EXAMPLE 6 | Calculate $\int_0^1 \tan^{-1}x\,dx$.

SOLUTION Let

$$u = \tan^{-1}x \qquad dv = dx$$

Then

$$du = \frac{dx}{1+x^2} \qquad v = x$$

So Formula 6 gives

$$\int_0^1 \tan^{-1}x\,dx = x\tan^{-1}x\Big]_0^1 - \int_0^1 \frac{x}{1+x^2}\,dx$$

$$= 1 \cdot \tan^{-1}1 - 0 \cdot \tan^{-1}0 - \int_0^1 \frac{x}{1+x^2}\,dx$$

$$= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2}\,dx$$

Since $\tan^{-1}x \geqslant 0$ for $x \geqslant 0$, the integral in Example 6 can be interpreted as the area of the region shown in Figure 2.

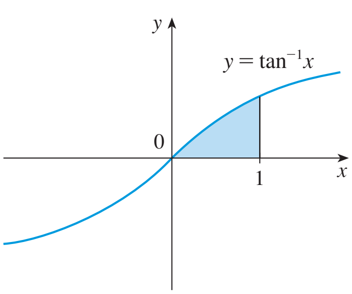


FIGURE 2

To evaluate this integral we use the substitution $t = 1 + x^2$ (since u has another meaning in this example). Then $dt = 2x\,dx$, so $x\,dx = \frac{1}{2}\,dt$. When $x = 0$, $t = 1$; when $x = 1$, $t = 2$; so

$$\int_0^1 \frac{x}{1+x^2}\,dx = \tfrac{1}{2}\int_1^2 \frac{dt}{t} = \tfrac{1}{2}\ln|t|\Big]_1^2$$

$$= \tfrac{1}{2}(\ln 2 - \ln 1) = \tfrac{1}{2}\ln 2$$

Therefore

$$\int_0^1 \tan^{-1}x\,dx = \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2}\,dx = \frac{\pi}{4} - \frac{\ln 2}{2}$$ ■

EXERCISES 5.5

1–2 Evaluate the integral using integration by parts with the indicated choices of u and dv.

1. $\int x^2 \ln x\, dx$; $u = \ln x$, $dv = x^2\, dx$

2. $\int \theta \cos\theta\, d\theta$; $u = \theta$, $dv = \cos\theta\, d\theta$

3–20 Evaluate the integral.

3. $\int x \cos 5x\, dx$

4. $\int xe^{-x}\, dx$

5. $\int re^{r/2}\, dr$

6. $\int t \sin 2t\, dt$

7. $\int x^2 \sin \pi x\, dx$

8. $\int x^2 \cos mx\, dx$

9. $\int \ln \sqrt[3]{x}\, dx$

10. $\int p^5 \ln p\, dp$

11. $\int e^{2\theta} \sin 3\theta\, d\theta$

12. $\int e^{-\theta} \cos 2\theta\, d\theta$

13. $\int_0^{\pi} t \sin 3t\, dt$

14. $\int_0^1 (x^2 + 1)e^{-x}\, dx$

15. $\int_1^2 \frac{\ln x}{x^2}\, dx$

16. $\int_4^9 \frac{\ln y}{\sqrt{y}}\, dy$

17. $\int_0^1 \frac{y}{e^{2y}}\, dy$

18. $\int_1^{\sqrt{3}} \arctan(1/x)\, dx$

19. $\int_1^2 (\ln x)^2\, dx$

20. $\int_0^1 \frac{r^3}{\sqrt{4 + r^2}}\, dr$

21–26 First make a substitution and then use integration by parts to evaluate the integral.

21. $\int \cos \sqrt{x}\, dx$

22. $\int t^3 e^{-t^2}\, dt$

23. $\int_{\sqrt{\pi/2}}^{\sqrt{\pi}} \theta^3 \cos(\theta^2)\, d\theta$

24. $\int_0^{\pi} e^{\cos t} \sin 2t\, dt$

25. $\int x \ln(1 + x)\, dx$

26. $\int \sin(\ln x)\, dx$

27. (a) If $n \geqslant 2$ is an integer, show that

$$\int \sin^n x\, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x\, dx$$

This is called a *reduction formula* because the exponent n has been *reduced* to $n - 1$ and $n - 2$.

(b) Use the reduction formula in part (a) to show that

$$\int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

(c) Use parts (a) and (b) to evaluate $\int \sin^4 x\, dx$.

28. (a) Prove the reduction formula

$$\int \cos^n x\, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x\, dx$$

(b) Use part (a) to evaluate $\int \cos^2 x\, dx$.
(c) Use parts (a) and (b) to evaluate $\int \cos^4 x\, dx$.

29–30 Use integration by parts to prove the reduction formula.

29. $\int (\ln x)^n\, dx = x(\ln x)^n - n \int (\ln x)^{n-1}\, dx$

30. $\int x^n e^x\, dx = x^n e^x - n \int x^{n-1} e^x\, dx$

31. Use Exercise 29 to find $\int (\ln x)^3\, dx$.

32. Use Exercise 30 to find $\int x^4 e^x\, dx$.

33. **Salicylic acid pharmacokinetics** In the article cited in Example 5 the authors also studied the formation and concentration of salicylic acid in the bloodstream of 10 volunteers. A model for the concentration is

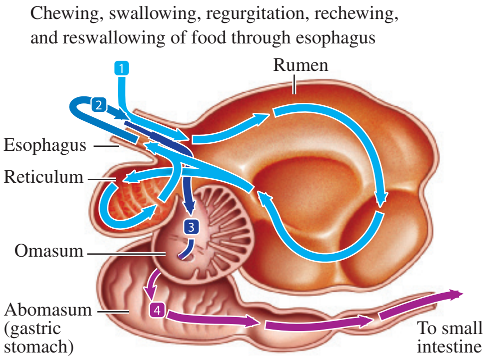
$$C(t) = 11.4te^{-t}$$

where t is measured in hours and C in μg/mL. Calculate $\int_0^4 C(t)\, dt$ and include the units in your answer.

34. **Rumen microbial ecosystem** The rumen is the first chamber in the stomach of ruminants such as cattle, sheep, and deer. Fermentation reactions by symbiotic organisms begin digesting plant matter in the rumen. If μ is the fraction of matter entering or leaving the rumen per unit time in a model for continuous fermentation, then the integral

$$\int_0^1 \mu e^{-\mu t}(1 - t)\, dt$$

is the fraction of soluble material passing from the rumen in the first hour without being fermented. Evaluate this integral.



Source: Adapted from R. E. Hungate, "The Rumen Microbial Ecosystem," *Annual Review of Ecology and Systematics* 6 (1975): 39–66.

35. Gene regulation In Section 10.3 a model of gene regulation is analyzed and it is shown that the concentration of protein in a cell as a function of time is given by the equation

$$p(t) = \tfrac{1}{2} - \tfrac{1}{2}e^{-t}(\sin t + \cos t)$$

The bioavailability of this protein is defined as the integral of this concentration over time. What is the bioavailability of the protein over the first unit of time?

36. Insect metamorphosis The rate of development of many insects increases gradually with temperature up to a maximum and then rapidly falls to zero. This can be approximated by the function $T - T^k$, where k is a positive integer and T is a standardized measure of temperature such that $0 \leqslant T \leqslant 1$ and $T = 0$ results in no development, whereas $T = 1$ is a lethal temperature. Suppose the daily temperature oscillates according to the equation $T(t) = a\cos^2 t$, where π units of time equals one day, and a is the factor between 0 and 1 by which the temperature oscillates over the course of a day. Use Exercise 28 to determine the cumulative amount of development that happens over the first week for the case where $k = 3$.

37. Suppose that $f(1) = 2$, $f(4) = 7$, $f'(1) = 5$, $f'(4) = 3$, and f'' is continuous. Find the value of $\int_1^4 x f''(x)\,dx$.

38. If $f(0) = g(0) = 0$ and f'' and g'' are continuous, show that

$$\int_0^a f(x)g''(x)\,dx = f(a)g'(a) - f'(a)g(a) + \int_0^a f''(x)g(x)\,dx$$

5.6 Partial Fractions

In Chapter 7 we will study the logistic model for population growth and in solving the logistic differential equation we will need to be able to integrate the function

$$F(N) = \frac{K}{N(K - N)}$$

where K is a positive constant. The integration techniques we have learned so far don't enable us to integrate this function, but in this section we learn how to do so. (See Example 2.)

The idea is to express a rational function as a sum of simpler functions, called *partial fractions*, that we already know how to integrate. To illustrate the method, observe that by taking the fractions $2/(x - 1)$ and $1/(x + 2)$ to a common denominator we obtain

$$\frac{2}{x-1} - \frac{1}{x+2} = \frac{2(x+2) - (x-1)}{(x-1)(x+2)} = \frac{x+5}{x^2 + x - 2}$$

If we now reverse the procedure, we see how to integrate the function on the right side of this equation:

$$\int \frac{x+5}{x^2+x-2}\,dx = \int \left(\frac{2}{x-1} - \frac{1}{x+2}\right) dx$$
$$= 2\ln|x-1| - \ln|x+2| + C$$

More generally, let's consider a rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials. It's possible to express f as a sum of simpler fractions provided that the degree of P is less than the degree of Q. If that's not the case, then we must take the preliminary step of dividing Q into P (by long division) until a remainder $R(x)$ is obtained whose degree is less than the degree of Q. As the following example illustrates, sometimes this preliminary step is all that is required.

EXAMPLE 1 | Find $\displaystyle\int \frac{x^3 + x}{x - 1}\,dx$.

SOLUTION Since the degree of the numerator is greater than the degree of the

$$\begin{array}{r}
x^2 + x + 2 \\
x - 1 \overline{)\, x^3 \qquad + x} \\
\underline{x^3 - x^2} \qquad \\
x^2 + x \\
\underline{x^2 - x} \\
2x \\
\underline{2x - 2} \\
2
\end{array}$$

denominator, we first perform the long division. This enables us to write

$$\int \frac{x^3 + x}{x - 1}\,dx = \int \left(x^2 + x + 2 + \frac{2}{x - 1} \right) dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln|x - 1| + C \qquad \blacksquare$$

The next step is to factor the denominator $Q(x)$ as far as possible. We concentrate on the case where $Q(x)$ is a product of distinct linear factors of the form $ax + b$. (Other cases are explored in the exercises.)

Then we express the function $R(x)/Q(x)$ as a sum of **partial fractions** of the form

$$\frac{A}{ax + b}$$

Each factor $ax + b$ of $Q(x)$ has a corresponding partial fraction. The constants in the numerators can be determined as in the following examples.

EXAMPLE 2 | **BB** Logistic model If K is a constant, evaluate the integral

$$\int \frac{K}{N(K - N)}$$

SOLUTION Because there are two linear factors, we write

$$\textbf{(1)} \qquad \frac{K}{N(K - N)} = \frac{A}{N} + \frac{B}{K - N}$$

where A and B are constants. To determine the values of A and B, we multiply both sides of Equation 1 by the product of denominators, $N(K - N)$:

$$K = A(K - N) + BN$$

$$K = (B - A)N + AK$$

Comparing the left and right sides, we see that this equation is satisfied if $B - A = 0$ and $A = 1$. So the solution is $A = B = 1$ and we have

$$\frac{K}{N(K - N)} = \frac{1}{N} + \frac{1}{K - N}$$

This new form of the function allows us to integrate more easily:

Notice that in integrating the second term we made the mental substitution $u = K - N$ and so $du = -dN$.

$$\int \frac{K}{N(K - N)}\,dN = \int \left(\frac{1}{N} + \frac{1}{K - N} \right) dN$$

$$= \ln|N| - \ln|K - N| + C \qquad \blacksquare$$

EXAMPLE 3 | Evaluate $\displaystyle\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x}\,dx$.

SOLUTION Since the degree of the numerator is less than the degree of the denominator, we don't need to divide. We factor the denominator as

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2)$$

Since the denominator has three distinct linear factors, the partial fraction decomposition of the integrand has the form

$$\frac{x^2 + 2x - 1}{x(2x-1)(x+2)} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2} \tag{2}$$

Another method for finding A, B, and C is given in the note after this example.

To determine the values of A, B, and C, we multiply both sides of this equation by the product of the denominators, $x(2x - 1)(x + 2)$, obtaining

$$x^2 + 2x - 1 = A(2x-1)(x+2) + Bx(x+2) + Cx(2x-1) \tag{3}$$

Expanding the right side of Equation 3 and writing it in the standard form for polynomials, we get

$$x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A \tag{4}$$

The polynomials in Equation 4 are identical, so their coefficients must be equal. The coefficient of x^2 on the right side, $2A + B + 2C$, must equal the coefficient of x^2 on the left side—namely, 1. Likewise, the coefficients of x are equal and the constant terms are equal. This gives the following system of equations for A, B, and C:

$$\begin{aligned} 2A + B + 2C &= 1 \\ 3A + 2B - C &= 2 \\ -2A &= -1 \end{aligned}$$

Solving, we get $A = \frac{1}{2}$, $B = \frac{1}{5}$, and $C = -\frac{1}{10}$, and so

We could check our work by taking the terms to a common denominator and adding them.

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x}\,dx = \int \left[\frac{1}{2}\,\frac{1}{x} + \frac{1}{5}\,\frac{1}{2x-1} - \frac{1}{10}\,\frac{1}{x+2}\right] dx$$

$$= \tfrac{1}{2}\ln|x| + \tfrac{1}{10}\ln|2x-1| - \tfrac{1}{10}\ln|x+2| + K$$

Figure 1 shows the graphs of the integrand in Example 3 and its indefinite integral (with $K = 0$). Which is which?

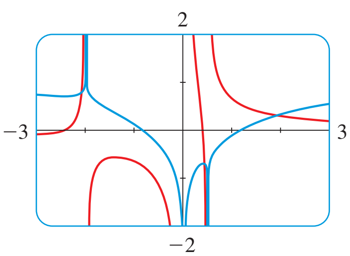


FIGURE 1

In integrating the middle term we have made the mental substitution $u = 2x - 1$, which gives $du = 2\,dx$ and $dx = \frac{1}{2}\,du$. ■

NOTE We can use an alternative method to find the coefficients A, B, and C in Example 3. Equation 3 is an identity; it is true for every value of x. Let's choose values of x that simplify the equation. If we put $x = 0$ in Equation 3, then the second and third terms on the right side vanish and the equation then becomes $-2A = -1$, or $A = \frac{1}{2}$. Likewise, $x = \frac{1}{2}$ gives $5B/4 = \frac{1}{4}$ and $x = -2$ gives $10C = -1$, so $B = \frac{1}{5}$ and $C = -\frac{1}{10}$. (You may object that Equation 2 is not valid for $x = 0, \frac{1}{2}$, or -2, so why should Equation 3 be valid for those values? In fact, Equation 4 is true for all values of x, even $x = 0, \frac{1}{2}$, and -2. See Exercise 23 for the reason.)

EXERCISES 5.6

1–2 Write the function as a sum of partial fractions. Do not determine the numerical values of the coefficients.

1. (a) $\dfrac{1}{x^2 - 1}$ (b) $\dfrac{2}{x^2 + x}$

2. (a) $\dfrac{x}{x^2 + x - 2}$ (b) $\dfrac{2 - x}{x^2 - 2x - 8}$

3–14 Evaluate the integral.

3. $\displaystyle\int \frac{x}{x-6}\,dx$

4. $\displaystyle\int \frac{r^2}{r+4}\,dr$

5. $\displaystyle\int \frac{x-9}{(x+5)(x-2)}\,dx$

6. $\displaystyle\int \frac{1}{(t+4)(t-1)}\,dt$

7. $\displaystyle\int_2^3 \frac{1}{x^2-1}\,dx$

8. $\displaystyle\int_0^1 \frac{x-1}{x^2+3x+2}\,dx$

9. $\displaystyle\int \frac{ax}{x^2 - bx}\,dx$

10. $\displaystyle\int \frac{1}{(x + a)(x + b)}\,dx$

11. $\displaystyle\int_0^1 \frac{2}{2x^2 + 3x + 1}\,dx$

12. $\displaystyle\int_0^1 \frac{x^3 - 4x - 10}{x^2 - x - 6}\,dx$

13. $\displaystyle\int_1^2 \frac{4y^2 - 7y - 12}{y(y + 2)(y - 3)}\,dy$

14. $\displaystyle\int \frac{x^2 + 2x - 1}{x^3 - x}\,dx$

15–18 Make a substitution to express the integrand as a rational function and then evaluate the integral.

15. $\displaystyle\int_9^{16} \frac{\sqrt{x}}{x - 4}\,dx$

16. $\displaystyle\int \frac{dx}{2\sqrt{x + 3} + x}$

17. $\displaystyle\int \frac{e^{2x}}{e^{2x} + 3e^x + 2}\,dx$

18. $\displaystyle\int \frac{\cos x}{\sin^2 x + \sin x}\,dx$

19. If a linear factor in the denominator of a rational function is repeated, there will be two corresponding partial fractions. For instance,

$$f(x) = \frac{5x^2 + 3x - 2}{x^2(x + 2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 2}$$

Determine the values of A, B, and C and use them to evaluate $\int f(x)\,dx$.

20. Use the method of Exercise 19 to evaluate

$$\int \frac{x^2 - 5x + 16}{(2x + 1)(x - 2)^2}\,dx$$

21. If a factor of the denominator is an irreducible quadratic, such as $x^2 + 1$, the corresponding partial fraction has a linear numerator. For instance,

$$f(x) = \frac{2x^2 + x + 1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

Determine the values of A, B, and C and use them to evaluate $\int f(x)\,dx$.

22. Use the method of Exercise 21 to evaluate

$$\int \frac{3x^2 - 2x + 3}{(x - 1)(x^2 + 1)}\,dx$$

23. Suppose that F, G, and Q are polynomials and

$$\frac{F(x)}{Q(x)} = \frac{G(x)}{Q(x)}$$

for all x except when $Q(x) = 0$. Prove that $F(x) = G(x)$ for all x. [*Hint:* Use continuity.]

BB **24. Sterile insect technique** One method of slowing the growth of an insect population without using pesticides is to introduce into the population a number of sterile males that mate with fertile females but produce no offspring. (The photo shows a screw-worm fly, the first pest effectively eliminated from a region by this method.) Let P represent the number of female insects in a population and S the number of sterile males introduced each generation. Let r be the per capita rate of production of females by females, provided their chosen mate is not sterile. Then the female population is related to time t by

$$t = \int \frac{P + S}{P[(r - 1)P - S]}\,dP$$

Suppose an insect population with 10,000 females grows at a rate of $r = 1.1$ and 900 sterile males are added. Evaluate the integral to give an equation relating the female population to time. (Note that the resulting equation can't be solved explicitly for P.)

USDA

5.7 Integration Using Tables and Computer Algebra Systems

We have not given an exhaustive treatment of techniques of integration, so it's important to know that there are resources available to scientists and others who encounter a difficult integral. Here we describe how to evaluate integrals using tables and computer algebra systems. For a *definite* integral that seems intractable, one can always approximate it by using the Midpoint Rule or the numerical integration capability of a scientific calculator.

Tables of Integrals

Tables of indefinite integrals are very useful when we are confronted by an integral that is difficult to evaluate by hand and we don't have access to a computer algebra system. A relatively brief table of 120 integrals, categorized by form, is provided on the Reference Pages at the front of the book. More extensive tables are available in the *CRC Standard Mathematical Tables and Formulae*, 32nd ed. by Daniel Zwillinger (Boca Raton, FL, 2011) (709 entries) or in Gradshteyn and Ryzhik's *Table of Integrals, Series, and Products*, 7e (San Diego, 2007), which contains hundreds of pages of integrals. It should be remembered, however, that integrals do not often occur in exactly the form listed in a table. Usually we need to use the Substitution Rule or algebraic manipulation to transform a given integral into one of the forms in the table.

EXAMPLE 1 | Use the Table of Integrals to evaluate $\int_0^2 \frac{x^2+12}{x^2+4}\,dx$.

The Table of Integrals appears on Reference Pages 6–10 at the front of the book.

SOLUTION The only formula in the table that resembles our given integral is entry 17:

$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\frac{u}{a} + C$$

If we perform long division, we get

$$\frac{x^2+12}{x^2+4} = 1 + \frac{8}{x^2+4}$$

Now we can use Formula 17 with $a = 2$:

$$\int_0^2 \frac{x^2+12}{x^2+4}\,dx = \int_0^2 \left(1 + \frac{8}{x^2+4}\right) dx = x + 8 \cdot \tfrac{1}{2}\tan^{-1}\frac{x}{2}\bigg]_0^2$$

$$= 2 + 4\tan^{-1} 1 = 2 + \pi$$

■

EXAMPLE 2 | Use the Table of Integrals to find $\int x^3 \sin x\,dx$.

SOLUTION If we look in the section called *Trigonometric Forms*, we see that none of the entries explicitly includes a u^3 factor. However, we can use the reduction formula in entry 84 with $n = 3$:

$$\int x^3 \sin x\,dx = -x^3\cos x + 3\int x^2 \cos x\,dx$$

85. $\int u^n \cos u\,du$
$= u^n \sin u - n\int u^{n-1}\sin u\,du$

We now need to evaluate $\int x^2 \cos x\,dx$. We can use the reduction formula in entry 85 with $n = 2$, followed by entry 82:

$$\int x^2\cos x\,dx = x^2\sin x - 2\int x\sin x\,dx$$

$$= x^2\sin x - 2(\sin x - x\cos x) + K$$

Combining these calculations, we get

$$\int x^3\sin x\,dx = -x^3\cos x + 3x^2\sin x + 6x\cos x - 6\sin x + C$$

where $C = 3K$.

■

EXAMPLE 3 | Use the Table of Integrals to find $\int x\sqrt{x^2+2x+4}\,dx$.

SOLUTION Since the table gives forms involving $\sqrt{a^2+x^2}$, $\sqrt{a^2-x^2}$, and $\sqrt{x^2-a^2}$, but not $\sqrt{ax^2+bx+c}$, we first complete the square:

$$x^2+2x+4=(x+1)^2+3$$

If we make the substitution $u=x+1$ (so $x=u-1$), the integrand will involve the pattern $\sqrt{a^2+u^2}$:

$$\begin{aligned}\int x\sqrt{x^2+2x+4}\,dx &= \int (u-1)\sqrt{u^2+3}\,du\\ &= \int u\sqrt{u^2+3}\,du - \int \sqrt{u^2+3}\,du\end{aligned}$$

The first integral is evaluated using the substitution $t=u^2+3$:

$$\int u\sqrt{u^2+3}\,du = \tfrac{1}{2}\int \sqrt{t}\,dt = \tfrac{1}{2}\cdot\tfrac{2}{3}t^{3/2} = \tfrac{1}{3}(u^2+3)^{3/2}$$

21. $\displaystyle\int \sqrt{a^2+u^2}\,du = \frac{u}{2}\sqrt{a^2+u^2} + \frac{a^2}{2}\ln\left(u+\sqrt{a^2+u^2}\right)+C$

For the second integral we use Formula 21 with $a=\sqrt{3}$:

$$\int \sqrt{u^2+3}\,du = \frac{u}{2}\sqrt{u^2+3} + \tfrac{3}{2}\ln\left(u+\sqrt{u^2+3}\right)$$

Thus

$$\begin{aligned}&\int x\sqrt{x^2+2x+4}\,dx\\ &= \tfrac{1}{3}(x^2+2x+4)^{3/2} - \frac{x+1}{2}\sqrt{x^2+2x+4} - \tfrac{3}{2}\ln\left(x+1+\sqrt{x^2+2x+4}\right)+C\end{aligned}$$

Computer Algebra Systems

We have seen that the use of tables involves matching the form of the given integrand with the forms of the integrands in the tables. Computers are particularly good at matching patterns. And just as we used substitutions in conjunction with tables, a CAS can perform substitutions that transform a given integral into one that occurs in its stored formulas. So it isn't surprising that computer algebra systems excel at integration. That doesn't mean that integration by hand is an obsolete skill. We will see that a hand computation sometimes produces an indefinite integral in a form that is more convenient than a machine answer.

To begin, let's see what happens when we ask a machine to integrate the relatively simple function $y=1/(3x-2)$. Using the substitution $u=3x-2$, an easy calculation by hand gives

$$\int \frac{1}{3x-2}\,dx = \tfrac{1}{3}\ln|3x-2|+C$$

whereas Derive, Mathematica, and Maple all return the answer

$$\tfrac{1}{3}\ln(3x-2)$$

The first thing to notice is that computer algebra systems omit the constant of integration. In other words, they produce a *particular* antiderivative, not the most general one. Therefore, when making use of a machine integration, we might have to add a constant.

Second, the absolute value signs are omitted in the machine answer. That is fine if our problem is concerned only with values of x greater than $\frac{2}{3}$. But if we are interested in other values of x, then we need to insert the absolute value symbol.

EXAMPLE 4 | Use a CAS to evaluate $\int x(x^2 + 5)^8\,dx$.

SOLUTION Maple and Mathematica give the same answer:

$$\tfrac{1}{18}x^{18} + \tfrac{5}{2}x^{16} + 50x^{14} + \tfrac{1750}{3}x^{12} + 4375x^{10} + 21875x^8 + \tfrac{218750}{3}x^6 + 156250x^4 + \tfrac{390625}{2}x^2$$

It's clear that both systems must have expanded $(x^2 + 5)^8$ by the Binomial Theorem and then integrated each term.

If we integrate by hand instead, using the substitution $u = x^2 + 5$, we get

Derive and the TI-89 and TI-92 also give this answer.

$$\int x(x^2 + 5)^8\,dx = \tfrac{1}{18}(x^2 + 5)^9 + C$$

For most purposes, this is a more convenient form of the answer. ■

EXAMPLE 5 | Use a CAS to find $\int \sin^5 x \cos^2 x\,dx$.

SOLUTION Derive and Maple report the answer

$$-\tfrac{1}{7}\sin^4 x \cos^3 x - \tfrac{4}{35}\sin^2 x \cos^3 x - \tfrac{8}{105}\cos^3 x$$

whereas Mathematica produces

$$-\tfrac{5}{64}\cos x - \tfrac{1}{192}\cos 3x + \tfrac{3}{320}\cos 5x - \tfrac{1}{448}\cos 7x$$

We suspect that there are trigonometric identities that we could use to show that these answers are equivalent. Indeed, if we ask Derive, Maple, and Mathematica to simplify their expressions using trigonometric identities, they ultimately produce the same form of the answer:

$$\int \sin^5 x \cos^2 x\,dx = -\tfrac{1}{3}\cos^3 x + \tfrac{2}{5}\cos^5 x - \tfrac{1}{7}\cos^7 x$$ ■

■ Can We Integrate All Continuous Functions?

The question arises: Will our basic integration formulas, together with the Substitution Rule, integration by parts, tables of integrals, and computer algebra systems, enable us to find the integral of every continuous function? In particular, can we use these techniques to evaluate $\int e^{x^2}\,dx$? The answer is No, at least not in terms of the functions that we are familiar with.

Most of the functions that we have been dealing with in this book are called **elementary functions**. These are the polynomials, rational functions, power functions (x^a), exponential functions (a^x), logarithmic functions, trigonometric and inverse trigonometric functions, and all functions that can be obtained from these by the five operations of addition, subtraction, multiplication, division, and composition. For instance, the function

$$f(x) = \sqrt{\frac{x^2 - 1}{x^3 + 2x - 1}} + \ln(\cos x) - xe^{\sin 2x}$$

is an elementary function.

If f is an elementary function, then f' is an elementary function but $\int f(x)\,dx$ need not be an elementary function. Consider $f(x) = e^{x^2}$. Since f is continuous, its integral exists, and if we define the function F by

$$F(x) = \int_0^x e^{t^2}\,dt$$

then we know from Part 1 of the Fundamental Theorem of Calculus that

$$F'(x) = e^{x^2}$$

Thus $f(x) = e^{x^2}$ has an antiderivative F, but it has been proved that F is not an elementary function. This means that no matter how hard we try, we will never succeed in evaluating $\int e^{x^2}\,dx$ in terms of the functions we know. The same can be said of the following integrals:

$$\int \frac{e^x}{x}\,dx \qquad \int \sin(x^2)\,dx \qquad \int \cos(e^x)\,dx$$

$$\int \sqrt{x^3+1}\,dx \qquad \int \frac{1}{\ln x}\,dx \qquad \int \frac{\sin x}{x}\,dx$$

In fact, the majority of elementary functions don't have elementary antiderivatives.

EXERCISES 5.7

1–18 Use the Table of Integrals on Reference Pages 6–10 to evaluate the integral.

1. $\int \tan^3(\pi x)\,dx$

2. $\int e^{2\theta}\sin 3\theta\,d\theta$

3. $\int \frac{dx}{x^2\sqrt{4x^2+9}}$

4. $\int_2^3 \frac{1}{x^2\sqrt{4x^2-7}}\,dx$

5. $\int e^{2x}\arctan(e^x)\,dx$

6. $\int \frac{\sqrt{2y^2-3}}{y^2}\,dy$

7. $\int_0^\pi x^3\sin x\,dx$

8. $\int \frac{dx}{2x^3-3x^2}$

9. $\int \frac{\tan^3(1/z)}{z^2}\,dz$

10. $\int x\sin(x^2)\cos(3x^2)\,dx$

11. $\int \sin^2 x\cos x\ln(\sin x)\,dx$

12. $\int \frac{\sin 2\theta}{\sqrt{5-\sin\theta}}\,d\theta$

13. $\int \frac{e^x}{3-e^{2x}}\,dx$

14. $\int_0^1 x^4e^{-x}\,dx$

15. $\int \frac{x^4\,dx}{\sqrt{x^{10}-2}}$

16. $\int \frac{\sec^2\theta\tan^2\theta}{\sqrt{9-\tan^2\theta}}\,d\theta$

17. $\int \frac{\sqrt{4+(\ln x)^2}}{x}\,dx$

18. $\int e^t\sin(\alpha t-3)\,dt$

19. Verify Formula 53 in the Table of Integrals (a) by differentiation and (b) by using the substitution $t = a + bu$.

20. Verify Formula 31 (a) by differentiation and (b) by substituting $u = a\sin\theta$.

CAS **21–27** Use a computer algebra system to evaluate the integral. Compare the answer with the result of using tables. If the answers are not the same, show that they are equivalent.

21. $\int \sec^4 x\,dx$

22. $\int x^2(1+x^3)^4\,dx$

23. $\int x\sqrt{1+2x}\,dx$

24. $\int \frac{dx}{e^x(3e^x+2)}$

25. $\int \tan^5 x\,dx$

26. $\int \sin^4 x\,dx$

27. $\int \frac{1}{\sqrt{1+\sqrt[3]{x}}}\,dx$

CAS **28.** Computer algebra systems sometimes need a helping hand from human beings. Try to evaluate

$$\int (1+\ln x)\sqrt{1+(x\ln x)^2}\,dx$$

with a computer algebra system. If it doesn't return an answer, make a substitution that changes the integral into one that the CAS *can* evaluate.

5.8 Improper Integrals

In defining a definite integral $\int_a^b f(x)\,dx$ we dealt with a function f defined on a finite interval $[a, b]$. In this section we extend the concept of a definite integral to the case where the interval is infinite. In this case the integral is called an *improper* integral. One of the most important applications of this idea, probability distributions, will be studied in Chapter 12.

Consider the infinite region S that lies under the curve $y = 1/x^2$, above the x-axis, and to the right of the line $x = 1$. You might think that, since S is infinite in extent, its area must be infinite, but let's take a closer look. The area of the part of S that lies to the left of the line $x = b$ (shaded in Figure 1) is

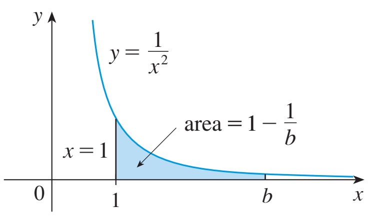


FIGURE 1

$$A(b) = \int_1^b \frac{1}{x^2}\,dx = -\frac{1}{x}\bigg]_1^b = 1 - \frac{1}{b}$$

Notice that $A(b) < 1$ no matter how large b is chosen.

We also observe that

$$\lim_{b\to\infty} A(b) = \lim_{b\to\infty}\left(1 - \frac{1}{b}\right) = 1$$

The area of the shaded region approaches 1 as $b \to \infty$ (see Figure 2), so we say that the area of the infinite region S is equal to 1 and we write

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\,dx = 1$$

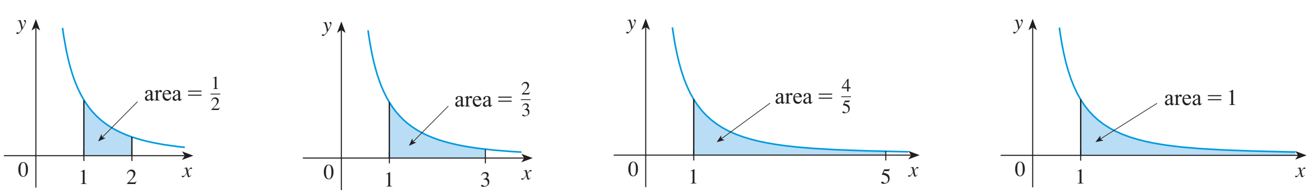


FIGURE 2

Using this example as a guide, we define the integral of f (not necessarily a positive function) over an infinite interval as the limit of integrals over finite intervals.

(1) Definition of an Improper Integral If $\int_a^b f(x)\,dx$ exists for every number $b \geqslant a$, then

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$$

provided this limit exists (as a finite number). An improper integral is called **convergent** if the corresponding limit exists and **divergent** if the limit does not exist.

The improper integral in Definition 1 can be interpreted as an area provided that f is a positive function. For instance, if $f(x) \geqslant 0$ and the integral $\int_a^\infty f(x)\,dx$ is convergent, then we define the area of the region $S = \{(x, y) \mid x \geqslant a, 0 \leqslant y \leqslant f(x)\}$ in Figure 3 to be

$$A(S) = \int_a^\infty f(x)\,dx$$

This is appropriate because $\int_a^\infty f(x)\,dx$ is the limit as $b \to \infty$ of the area under the graph of f from a to b.

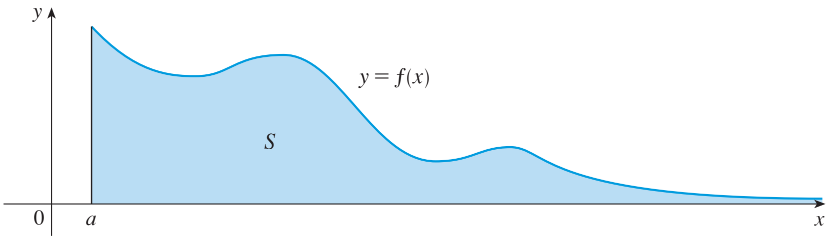


FIGURE 3

EXAMPLE 1 | Determine whether the integral $\int_1^\infty (1/x)\,dx$ is convergent or divergent.

SOLUTION According to Definition 1, we have

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx = \lim_{b\to\infty} \ln|x|\Big]_1^b$$

$$= \lim_{b\to\infty} (\ln b - \ln 1) = \lim_{b\to\infty} \ln b = \infty$$

The limit does not exist as a finite number and so the improper integral $\int_1^\infty (1/x)\,dx$ is divergent. ■

Let's compare the result of Example 1 with the example given at the beginning of this section:

$$\int_1^\infty \frac{1}{x^2}\,dx \;\text{ converges} \qquad\qquad \int_1^\infty \frac{1}{x}\,dx \;\text{ diverges}$$

Geometrically, this says that although the curves $y = 1/x^2$ and $y = 1/x$ look very similar for $x > 0$, the region under $y = 1/x^2$ to the right of $x = 1$ (the shaded region in Figure 4) has finite area whereas the corresponding region under $y = 1/x$ (in Figure 5) has infinite area. Note that both $1/x^2$ and $1/x$ approach 0 as $x \to \infty$ but $1/x^2$ approaches 0 faster than $1/x$. The values of $1/x$ don't decrease fast enough for its integral to have a finite value.

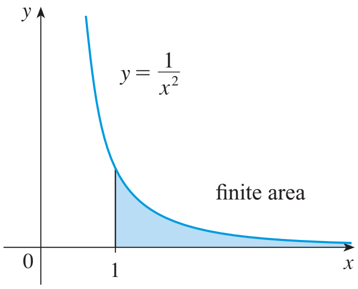


FIGURE 4

$\int_1^\infty (1/x^2)\,dx$ converges.

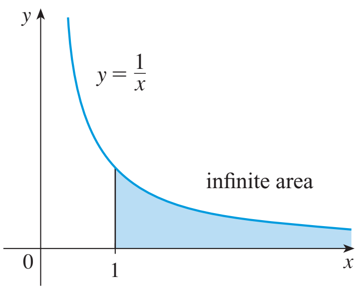


FIGURE 5

$\int_1^\infty (1/x)\,dx$ diverges.

EXAMPLE 2 | **Salicylic acid pharmacokinetics** In Exercises 5.2.14 and 5.5.33 we used the function $C(t) = 11.4te^{-t}$ to model the concentration of SA in the bloodstream of volunteers, where t is measured in hours and C is measured in μg/mL. Calculate $\int_0^\infty C(t)\,dt$ and interpret it.

SOLUTION The definition of an improper integral says that

$$\int_0^\infty C(t)\,dt = \lim_{b\to\infty}\int_0^b C(t)\,dt = 11.4\lim_{b\to\infty}\int_0^b te^{-t}\,dt$$

We integrate by parts with $u = t$, $dv = e^{-t}\,dt$, so $du = dt$ and $v = -e^{-t}$:

$$\int_0^b te^{-t}\,dt = -te^{-t}\Big]_0^b - \int_0^b (-e^{-t})\,dt$$

$$= -be^{-b} - 0 - e^{-t}\Big]_0^b = -be^{-b} - e^{-b} + 1$$

TEC In Module 5.8 you can investigate visually and numerically whether several improper integrals are convergent or divergent.

We know that $e^{-b} \to 0$ as $b \to \infty$, and by l'Hospital's Rule we have

$$\lim_{b\to\infty} be^{-b} = \lim_{b\to\infty}\frac{b}{e^b} = \lim_{b\to\infty}\frac{1}{e^b}$$

$$= \lim_{b\to\infty} e^{-b} = 0$$

Therefore

$$\int_0^\infty C(t)\,dt = 11.4\lim_{b\to\infty}(-be^{-b} - e^{-b} + 1)$$

$$= 11.4(-0 - 0 + 1) = 11.4\ (\mu\text{g/mL})\cdot\text{h}$$

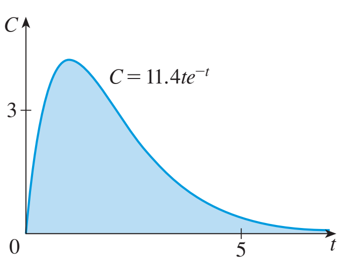


FIGURE 6
$\int_0^\infty C(t)\,dt = 11.4\ (\mu\text{g/mL})\cdot\text{h}$

We have previously interpreted $\int_0^1 C(t)\,dt$ in terms of the "availability" of the drug during the first hour, with units of concentration times time. Similarly, $\int_0^4 C(t)\,dt$ measures the availability over the first four hours. The improper integral $\int_0^\infty C(t)\,dt$ (the total area under the concentration curve in Figure 6) measures the availability for all time. In other words, it measures the long-term availability of SA. ■

We next define an improper integral over an interval that extends infinitely far in the negative direction.

(2) Definition If $\int_a^b f(x)\,dx$ exists for every number $a \leqslant b$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx$$

provided this limit exists (as a finite number).

EXAMPLE 3 | Evaluate $\int_{-\infty}^0 e^x\,dx$, if it is convergent.

SOLUTION Using Definition 2, we have

$$\int_{-\infty}^0 e^x\,dx = \lim_{a\to-\infty}\int_a^0 e^x\,dx = \lim_{a\to-\infty}\Big[e^x\Big]_a^0$$

$$= \lim_{a\to-\infty}(1 - e^a) = 1$$

because $\lim_{a\to-\infty} e^a = 0$. ■

Finally, we define an integral over the entire real line by splitting the real line into separate parts.

(3) Definition If both $\int_c^\infty f(x)\,dx$ and $\int_{-\infty}^c f(x)\,dx$ are convergent, then we define

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_{c}^{\infty} f(x)\,dx$$

where c is any real number. (See Exercise 30).

EXAMPLE 4 | Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$.

SOLUTION It's convenient to choose $c = 0$ in Definition 3:

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx$$

We must now evaluate the integrals on the right side separately:

$$\int_0^\infty \frac{1}{1+x^2}\,dx = \lim_{b\to\infty} \int_0^b \frac{dx}{1+x^2} = \lim_{b\to\infty} \tan^{-1}x\Big]_0^b$$
$$= \lim_{b\to\infty} (\tan^{-1}b - \tan^{-1}0) = \lim_{b\to\infty} \tan^{-1}b = \frac{\pi}{2}$$

$$\int_{-\infty}^0 \frac{1}{1+x^2}\,dx = \lim_{a\to-\infty} \int_a^0 \frac{dx}{1+x^2} = \lim_{a\to-\infty} \tan^{-1}x\Big]_a^0$$
$$= \lim_{a\to-\infty} (\tan^{-1}0 - \tan^{-1}a)$$
$$= 0 - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$$

Since both of these integrals are convergent, the given integral is convergent and

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

Since $1/(1+x^2) > 0$, the given improper integral can be interpreted as the area of the infinite region that lies under the curve $y = 1/(1+x^2)$ and above the x-axis (see Figure 7). ■

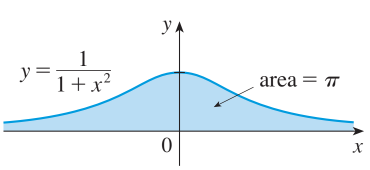


FIGURE 7

Improper integrals are used in various areas of biology, a few of which are illustrated in Example 2 and Exercises 25–28. In the study of continuous probability distributions many improper integrals arise. Probability density functions f have the property that

We will study continuous probability distributions in Chapter 12.

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

and a basic fact for normal distributions is that

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

(The proof of this fact requires methods that are beyond the scope of this book.)

Improper integrals also occur with great frequency in the mathematics required for medical imaging. In particular, functions defined in terms of improper integrals have become the basic tools needed to develop the theory of the CT scan, which creates images from X-ray data.

EXERCISES 5.8

1. Find the area under the curve $y = 1/x^3$ from $x = 1$ to $x = b$ and evaluate it for $b = 10$, 100, and 1000. Then find the total area under this curve for $x \geqslant 1$.

2. (a) Graph the functions $f(x) = 1/x^{1.1}$ and $g(x) = 1/x^{0.9}$ in the viewing rectangles $[0, 10]$ by $[0, 1]$ and $[0, 100]$ by $[0, 1]$.
(b) Find the areas under the graphs of f and g from $x = 1$ to $x = b$ and evaluate for $b = 10$, 100, 10^4, 10^6, 10^{10}, and 10^{20}.
(c) Find the total area under each curve for $x \geqslant 1$, if it exists.

3–22 Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

3. $\displaystyle\int_3^\infty \frac{1}{(x-2)^{3/2}}\,dx$

4. $\displaystyle\int_0^\infty \frac{1}{\sqrt[4]{1+x}}\,dx$

5. $\displaystyle\int_{-\infty}^{-1} \frac{1}{\sqrt{2-w}}\,dw$

6. $\displaystyle\int_0^\infty \frac{x}{(x^2+2)^2}\,dx$

7. $\displaystyle\int_4^\infty e^{-y/2}\,dy$

8. $\displaystyle\int_{-\infty}^{-1} e^{-2t}\,dt$

9. $\displaystyle\int_{2\pi}^\infty \sin\theta\,d\theta$

10. $\displaystyle\int_{-\infty}^\infty (y^3 - 3y^2)\,dy$

11. $\displaystyle\int_{-\infty}^\infty xe^{-x^2}\,dx$

12. $\displaystyle\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

13. $\displaystyle\int_1^\infty \frac{x+1}{x^2+2x}\,dx$

14. $\displaystyle\int_{-\infty}^\infty \cos\pi t\,dt$

15. $\displaystyle\int_0^\infty se^{-5s}\,ds$

16. $\displaystyle\int_{-\infty}^6 re^{r/3}\,dr$

17. $\displaystyle\int_1^\infty \frac{\ln x}{x}\,dx$

18. $\displaystyle\int_{-\infty}^\infty x^3e^{-x^4}\,dx$

19. $\displaystyle\int_{-\infty}^\infty \frac{x^2}{9+x^6}\,dx$

20. $\displaystyle\int_1^\infty \frac{\ln x}{x^3}\,dx$

21. $\displaystyle\int_e^\infty \frac{1}{x(\ln x)^3}\,dx$

22. $\displaystyle\int_0^\infty \frac{e^x}{e^{2x}+3}\,dx$

23–24 Sketch the region and find its area (if the area is finite).

23. $S = \{(x, y) \mid x \leqslant 1,\ 0 \leqslant y \leqslant e^x\}$

24. $S = \{(x, y) \mid x \geqslant -2,\ 0 \leqslant y \leqslant e^{-x/2}\}$

25. Drug pharmacokinetics The plasma drug concentration of a new drug was modeled by the function $C(t) = 23te^{-2t}$, where t is measured in hours and C in μg/mL.
(a) What is the maximum drug concentration and when did it occur?
(b) Calculate $\int_0^\infty C(t)\,dt$ and explain its significance.

26. Spread of drug use In a study of the spread of illicit drug use from an enthusiastic user to a population of N users, the authors model the number of expected new users by the equation

$$\gamma = \int_0^\infty \frac{cN(1 - e^{-kt})}{k} e^{-\lambda t}\,dt$$

where c, k, and λ are positive constants. Evaluate this integral to express γ in terms of c, N, k, and λ.

Source: Adapted from F. C. Hoppensteadt et al., "Threshold Analysis of a Drug Use Epidemic Model," *Mathematical Biosciences* 53 (1981): 79–87.

27. Photosynthesis Much of the earth's photosynthesis occurs in the oceans. The rate of primary production depends on light intensity, measured as the flux of photons (that is, number of photons per unit area per unit time). For monochromatic light, intensity decreases with water depth according to Beer's Law, which states that $I(x) = e^{-kx}$, where x is water depth. A simple model for the relationship between rate of photosynthesis and light intensity is $P(I) = aI$, where a is a constant and P is measured as a mass of carbon fixed per volume of water, per unit time. Calculate $\int_0^\infty P(I(x))\,dx$ and interpret it.

Source: Adapted from A. Jassby et al., "Mathematical Formulation of the Relationship between Photosynthesis and Light for Phytoplankton," *Limnology and Oceanography* 21 (1976): 540–7.

28. Dialysis treatment removes urea and other waste products from a patient's blood by diverting some of the bloodflow externally through a machine called a dialyzer. The rate at which urea is removed from the blood (in mg/min) is often well described by the equation

$$c(t) = \frac{K}{V} c_0 e^{-Kt/V}$$

where K is the rate of flow of blood through the dialyzer (in mL/min), V is the volume of the patient's blood (in mL), and c_0 is the amount of urea in the blood (in mg) at time $t = 0$. Evaluate the integral $\int_0^\infty c(t)\,dt$ and interpret it.

29. A manufacturer of lightbulbs wants to produce bulbs that last about 700 hours but, of course, some bulbs burn out faster than others. Let $F(t)$ be the fraction of the company's bulbs that burn out before t hours, so $F(t)$ always lies between 0 and 1.
(a) Make a rough sketch of what you think the graph of F might look like.
(b) What is the meaning of the derivative $r(t) = F'(t)$?
(c) What is the value of $\int_0^\infty r(t)\,dt$? Why?

30. If $\int_{-\infty}^\infty f(x)\,dx$ is convergent and a and b are real numbers, show that

$$\int_{-\infty}^a f(x)\,dx + \int_a^\infty f(x)\,dx = \int_{-\infty}^b f(x)\,dx + \int_b^\infty f(x)\,dx$$

31. (a) Show that $\int_{-\infty}^\infty x\,dx$ is divergent.
(b) Show that

$$\lim_{t\to\infty} \int_{-t}^t x\,dx = 0$$

This shows that we can't define

$$\int_{-\infty}^\infty f(x)\,dx = \lim_{t\to\infty} \int_{-t}^t f(x)\,dx$$

32. For what values of p is the integral

$$\int_1^\infty \frac{1}{x^p}\,dx$$

convergent? Evaluate the integral for those values of p.

33–35 Evaluate the integral, given that

$$\int_0^\infty e^{-x^2}\,dx = \tfrac{1}{2}\sqrt{\pi}$$

33. $\int_0^\infty e^{-x^2/2}\,dx$

34. $\int_0^\infty x^2 e^{-x^2}\,dx$

35. $\int_0^\infty \sqrt{x}\,e^{-x}\,dx$

Chapter 5 REVIEW

CONCEPT CHECK

1. (a) Write an expression for a Riemann sum of a function f. Explain the meaning of the notation that you use.
(b) If $f(x) \geqslant 0$, what is the geometric interpretation of a Riemann sum? Illustrate with a diagram.
(c) If $f(x)$ takes on both positive and negative values, what is the geometric interpretation of a Riemann sum? Illustrate with a diagram.

2. (a) Write the definition of the definite integral of a continuous function from a to b.
(b) What is the geometric interpretation of $\int_a^b f(x)\,dx$ if $f(x) \geqslant 0$?
(c) What is the geometric interpretation of $\int_a^b f(x)\,dx$ if $f(x)$ takes on both positive and negative values? Illustrate with a diagram.

3. State the Midpoint Rule.

4. (a) State the Evaluation Theorem.
(b) State the Net Change Theorem.

5. If $r(t)$ is the rate of growth of a population at time t, where t is measured in months, what does $\int_6^{10} r(t)\,dt$ represent?

6. (a) Explain the meaning of the indefinite integral $\int f(x)\,dx$.
(b) What is the connection between the definite integral $\int_a^b f(x)\,dx$ and the indefinite integral $\int f(x)\,dx$?

7. State both parts of the Fundamental Theorem of Calculus.

8. (a) State the Substitution Rule. In practice, how do you use it?
(b) State the rule for integration by parts. In practice, how do you use it?

9. Define the following improper integrals.
(a) $\int_a^\infty f(x)\,dx$ (b) $\int_{-\infty}^b f(x)\,dx$ (c) $\int_{-\infty}^\infty f(x)\,dx$

10. Explain exactly what is meant by the statement that "differentiation and integration are inverse processes."

Answers to the Concept Check can be found on the back endpapers.

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

1. If f and g are continuous on $[a, b]$, then

$$\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

2. If f and g are continuous on $[a, b]$, then

$$\int_a^b [f(x)g(x)]\,dx = \left(\int_a^b f(x)\,dx\right)\left(\int_a^b g(x)\,dx\right)$$

3. If f is continuous on $[a, b]$, then

$$\int_a^b 5f(x)\,dx = 5\int_a^b f(x)\,dx$$

4. If f is continuous on $[a, b]$, then

$$\int_a^b xf(x)\,dx = x\int_a^b f(x)\,dx$$

5. If f is continuous on $[a, b]$ and $f(x) \geqslant 0$, then

$$\int_a^b \sqrt{f(x)}\,dx = \sqrt{\int_a^b f(x)\,dx}$$

6. If f' is continuous on $[1, 3]$, then $\int_1^3 f'(v)\,dv = f(3) - f(1)$.

7. If f and g are continuous and $f(x) \geqslant g(x)$ for $a \leqslant x \leqslant b$, then

$$\int_a^b f(x)\,dx \geqslant \int_a^b g(x)\,dx$$

8. If f and g are differentiable and $f(x) \geqslant g(x)$ for $a < x < b$, then $f'(x) \geqslant g'(x)$ for $a < x < b$.

9. $\displaystyle\int_{-1}^1 \left(x^5 - 6x^9 + \frac{\sin x}{(1 + x^4)^2}\right) dx = 0$

10. $\displaystyle\int_{-5}^5 (ax^2 + bx + c)\,dx = 2\int_0^5 (ax^2 + c)\,dx$

11. $\int_0^2 (x - x^3)\,dx$ represents the area under the curve $y = x - x^3$ from 0 to 2.

12. All continuous functions have antiderivatives.

13. All continuous functions have derivatives.

14. If f is continuous, then $\int_{-\infty}^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_{-t}^{t} f(x)\,dx$.

15. If f is a continuous, decreasing function on $[1, \infty)$ and $\lim_{t\to\infty} f(x) = 0$, then $\int_1^\infty f(x)\,dx$ is convergent.

16. If $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)\,dx$ are both convergent, then $\int_a^\infty [f(x) + g(x)]\,dx$ is convergent.

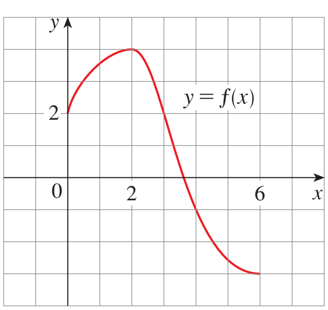
17. If $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)\,dx$ are both divergent, then $\int_a^\infty [f(x) + g(x)]\,dx$ is divergent.

18. If f is continuous on $[a, b]$, then

$$\frac{d}{dx}\left(\int_a^b f(x)\,dx\right) = f(x)$$

EXERCISES

1. Use the given graph of f to find the Riemann sum with six subintervals. Take the sample points to be (a) left endpoints and (b) midpoints. In each case draw a diagram and explain what the Riemann sum represents.



2. (a) Evaluate the Riemann sum for

$$f(x) = x^2 - x \qquad 0 \leqslant x \leqslant 2$$

with four subintervals, taking the sample points to be right endpoints. Explain, with the aid of a diagram, what the Riemann sum represents.

(b) Use the definition of a definite integral (with right endpoints) to calculate the value of the integral

$$\int_0^2 (x^2 - x)\,dx$$

(c) Use the Evaluation Theorem to check your answer to part (b).

(d) Draw a diagram to explain the geometric meaning of the integral in part (b).

3. Evaluate

$$\int_0^1 \left(x + \sqrt{1 - x^2}\right) dx$$

by interpreting it in terms of areas.

4. Express

$$\lim_{n\to\infty} \sum_{i=1}^{n} \sin x_i\,\Delta x$$

as a definite integral on the interval $[0, \pi]$ and then evaluate the integral.

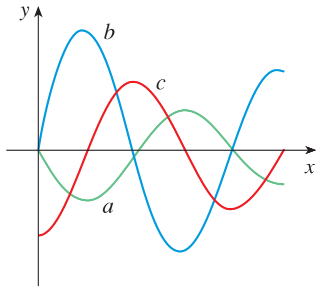
5. If $\int_0^6 f(x)\,dx = 10$ and $\int_0^4 f(x)\,dx = 7$, find $\int_4^6 f(x)\,dx$.

6. (a) Write $\int_0^3 e^{-x/2}\,dx$ as a limit of Riemann sums, taking the sample points to be right endpoints.

(b) Use the Midpoint Rule with six subintervals to estimate the value of the integral in part (a). State your answer correct to three decimal places.

(c) Use the Fundamental Theorem to evaluate $\int_0^3 e^{-x/2}\,dx$. Round your answer to three decimal places and compare with your estimate in part (b).

7. The following figure shows the graphs of f, f', and $\int_0^x f(t)\,dt$. Identify each graph, and explain your choices.



8. Evaluate:

(a) $\displaystyle\int_0^1 \frac{d}{dx}\left(e^{\arctan x}\right) dx$ (b) $\displaystyle\frac{d}{dx}\int_0^1 e^{\arctan x}\, dx$

(c) $\displaystyle\frac{d}{dx}\int_0^x e^{\arctan t}\, dt$

9–32 Evaluate the integral.

9. $\displaystyle\int_1^2 (8x^3 + 3x^2)\, dx$

10. $\displaystyle\int_0^T (x^4 - 8x + 7)\, dx$

11. $\displaystyle\int_0^1 (1 - x^9)\, dx$

12. $\displaystyle\int_0^1 (1 - x)^9\, dx$

13. $\displaystyle\int \left(\frac{1 - x}{x}\right)^2 dx$

14. $\displaystyle\int_0^1 (\sqrt[4]{u} + 1)^2\, du$

15. $\displaystyle\int_0^1 \frac{x}{x^2 + 1}\, dx$

16. $\displaystyle\int \frac{\csc^2 x}{1 + \cot x}\, dx$

17. $\displaystyle\int_0^1 v^2 \cos(v^3)\, dv$

18. $\displaystyle\int_0^1 \sin(3\pi t)\, dt$

19. $\displaystyle\int_0^1 e^{\pi t}\, dt$

20. $\displaystyle\int_1^2 \frac{1}{2 - 3x}\, dx$

21. $\displaystyle\int \frac{x + 2}{\sqrt{x^2 + 4x}}\, dx$

22. $\displaystyle\int_1^2 x^3 \ln x\, dx$

23. $\displaystyle\int_0^5 \frac{x}{x + 10}\, dx$

24. $\displaystyle\int_0^5 y e^{-0.6y}\, dy$

25. $\displaystyle\int_{-\pi/4}^{\pi/4} \frac{t^4 \tan t}{2 + \cos t}\, dt$

26. $\displaystyle\int_1^4 \frac{dt}{(2t + 1)^3}$

27. $\displaystyle\int_1^4 x^{3/2} \ln x\, dx$

28. $\displaystyle\int \sin x \cos(\cos x)\, dx$

29. $\displaystyle\int e^{\sqrt[3]{x}}\, dx$

30. $\displaystyle\int \tan^{-1} x\, dx$

31. $\displaystyle\int \frac{\sec\theta \tan\theta}{1 + \sec\theta}\, d\theta$

32. $\displaystyle\int_0^1 \frac{e^x}{1 + e^{2x}}\, dx$

33. Use a graph to give a rough estimate of the area of the region that lies under the curve $y = x\sqrt{x}$, $0 \leqslant x \leqslant 4$. Then find the exact area.

34–35 Find the derivative of the function.

34. $\displaystyle F(x) = \int_0^x \frac{t^2}{1 + t^3}\, dt$

35. $\displaystyle g(x) = \int_1^{\sin x} \frac{1 - t^2}{1 + t^4}\, dt$

36–38 Use the Table of Integrals on the Reference Pages to evaluate the integral.

36. $\displaystyle\int \csc^5 t\, dt$

37. $\displaystyle\int e^x \sqrt{1 - e^{2x}}\, dx$

38. $\displaystyle\int \frac{\cot x}{\sqrt{1 + 2\sin x}}\, dx$

39. Use Property 8 of integrals (page 338) to estimate the value of

$$\int_1^3 \sqrt{x^2 + 3}\, dx$$

40. Use the properties of integrals to verify that

$$0 \leqslant \int_0^1 x^4 \cos x\, dx \leqslant 0.2$$

41–43 Evaluate the integral or show that it is divergent.

41. $\displaystyle\int_1^\infty \frac{1}{(2x + 1)^3}\, dx$

42. $\displaystyle\int_0^\infty \frac{\ln x}{x^4}\, dx$

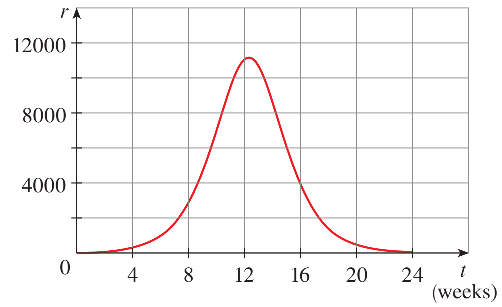
43. $\displaystyle\int_{-\infty}^0 e^{-2x}\, dx$

44. The speedometer reading v on a car was observed at one-minute intervals and recorded in the chart. Use the Midpoint Rule to estimate the distance traveled by the car.

t (min)	v (mi/h)	t (min)	v (mi/h)
0	40	6	56
1	42	7	57
2	45	8	57
3	49	9	55
4	52	10	56
5	54		

45. Let $r(t)$ be the rate at which the world's oil is consumed, where t is measured in years starting at $t = 0$ on January 1, 2000, and $r(t)$ is measured in barrels per year. What does $\int_0^{15} r(t)\, dt$ represent?

46. A **population of honeybees** increased at a rate of $r(t)$ bees per week, where the graph of r is as shown. Use the Midpoint Rule with six subintervals to estimate the increase in the bee population during the first 24 weeks.



47. An oil leak from a well is causing pollution at a rate of $r(t) = 90e^{-0.12t}$ gallons per month. If the leak is never fixed, what is the total amount of oil that will be spilled?

48. Antibiotic pharmacokinetics An antibiotic tablet is taken and t hours later the concentration in the bloodstream is

$$C(t) = 3(e^{-0.8t} - e^{-1.2t})$$

where C is measured in μg/mL.

(a) What is the maximum concentration of the antibiotic and when does it occur?

(b) Calculate $\int_0^2 C(t)\,dt$ and interpret your result.

(c) Calculate $\int_0^\infty C(t)\,dt$ and explain its meaning.

49. Population dynamics Suppose that the birth and death rates in a population change through time according to the functions $b(t)$ and $d(t)$. The net rate of change is defined as $r(t) = b(t) - d(t)$.

(a) Find an expression for the net change in population size between times $t = a$ and $t = b$ in terms of $r(t)$.

(b) Use Property 4 of integrals (page 336) to show that your answer to part (a) can also be expressed in terms of the total number of births and the total number of deaths over this period.

50. Angiotensin-converting enzyme inhibitors are medications that reduce blood pressure by dilating blood vessels. The rate of change of blood pressure with respect to dosage is given by the equation

$$P'(d) = -\frac{8\eta l v R'(d)}{R(d)^3}$$

where v is blood velocity, η is blood viscosity, l is the length of the blood vessel, and $R(d)$ is the radius of the vessel as a function of the dose d. Use a substitution to integrate $P'(d)$ and show that you obtain Poiseuille's Law:

$$P(d) = \frac{4\eta l v}{R(d)^2}$$

51. Environmental pollutants In Section 10.3 a model for the transport of environmental pollutants between three lakes is analyzed. It is shown that, for certain parameter values, the concentration of pollutant in one of the lakes as a function of time is given by an equation of the form

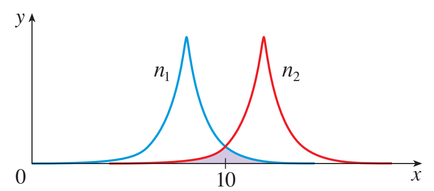
$$x(t) = k - ke^{-at}\cos bt$$

The environmental impact of the pollutant is a function of both its concentration and the duration of time that it persists. The integral of the concentration over time is a summary measure of this impact. Calculate this measure of impact over the first unit of time.

52. Niche overlap The extent to which species compete for resources is often measured by the *niche overlap*. If the horizontal axis represents a continuum of different resource types (for example, seed sizes for certain bird species), then a plot of the degree of preference for these resources is called a *species' niche*. The degree of overlap of two species' niches is then a measure of the extent to which they compete for resources. The niche overlap for a species is the fraction of the area under its preference curve that is also under the other species' curve. Many species' niches are best modeled by a function that has a peak at some intermediate resource type and decreases to 0 asymptotically. The niches displayed in the figure are given by

$$n_1(x) = e^{-|x-8|} \qquad n_2(x) = e^{-|x-12|}$$

for species 1 and 2, respectively. Calculate the niche overlap for species 1.



53. If f is a continuous function such that

$$\int_0^x f(t)\,dt = xe^{2x} + \int_0^x e^{-t} f(t)\,dt$$

for all x, find an explicit formula for $f(x)$.

54. Find a function f and a value of the constant a such that

$$2\int_a^x f(t)\,dt = 2\sin x - 1$$

55. If f' is continuous on $[a, b]$, show that

$$2\int_a^b f(x)f'(x)\,dx = [f(b)]^2 - [f(a)]^2$$

56. If f' is continuous on $[0, \infty)$ and $\lim_{x\to\infty} f(x) = 0$, show that

$$\int_0^\infty f'(x)\,dx = -f(0)$$

CASE STUDY 1c Kill Curves and Antibiotic Effectiveness

Recall that in this case study we are exploring the relationship between the magnitude of antibiotic treatment and the effectiveness of the treatment. One of the most important components of our analysis is the antibiotic concentration profile, which is a plot of the antibiotic concentration as a function of time.

In the simple model of Case Studies 1a and 1b we modeled a single dose of antibiotic using the equation

$$\textbf{(1)} \qquad \frac{dc}{dt} = -kc$$

for some positive constant k. From this we saw that the concentration as a function of time is

$$\textbf{(2)} \qquad c(t) = c_0 e^{-kt}$$

where c_0 is the concentration at $t = 0$. (See Figure 1.)

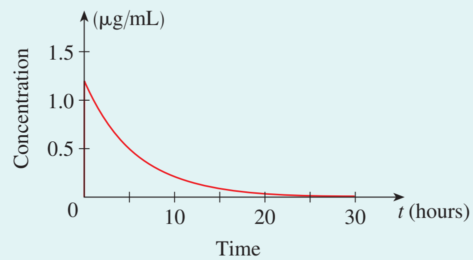


FIGURE 1
Antibiotic concentration profile modeled by the function $c(t) = c_0 e^{-kt}$ with $c_0 = 1.2\ \mu\text{g/mL}$ and $k = 0.175$

Three of the most common measures of the magnitude of antibiotic treatment are (1) the peak antibiotic concentration divided by *MIC*, denoted by ρ; (2) the duration of time for which the antibiotic concentration remains above *MIC*, denoted by τ; and (3) the area under the antibiotic concentration profile divided by *MIC*, denoted by α. These are shown in Figure 2. In Case Study 1a you derived expressions for the first two.

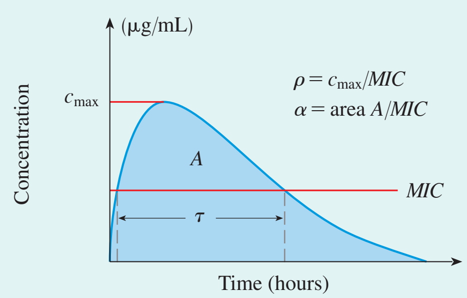


FIGURE 2
Three measures of the magnitude of antibiotic treatment

1. Find an expression for α in terms of k, c_0, and *MIC*, using Equation 2. The area under the concentration profile should be calculated from $t = 0$ to ∞.

We saw in Case Study 1a that, for a given antibiotic and bacteria species (in other words, for a given value of k and *MIC*), all three quantities ρ, τ, and α increase with one another. In other words, it is not possible to have a high value of α without also having high values of ρ and τ. Here you will show that we can break this dependency if we divide the total amount of antibiotic given c_0 into multiple smaller doses. This is referred to as *dose fractionation*.

In the simplest case, suppose that instead of giving a total amount of c_0 $\mu g/mL$ of antibiotic at $t = 0$, we instead give $c_0/2$ at $t = 0$ and another dose of $c_0/2$ at time $t = \hat{t}$. The time $\hat{t}$ is called the interdose interval. Furthermore, suppose that at each dose the concentration instantly increases by $c_0/2$, and otherwise it decays according to Equation 1.

2. Find an equation for the concentration as a function of time. Figure 3 plots this function for a specific choice of constants, along with the concentration profile when a single dose of c_0 $\mu g/mL$ is given at $t = 0$.

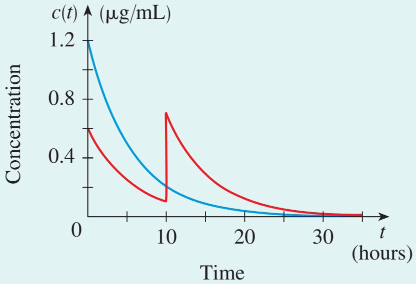


FIGURE 3
Red curve is the concentration profile modeled by the function from Problem 2 with $c_0 = 1.2$ $\mu g/mL$, $\hat{t} = 10$, and $k = 0.175$. Blue curve is the concentration profile when all the antibiotic is given at $t = 0$.

3. Use your answer to Problem 2 to find an expression for ρ, the peak concentration divided by *MIC*.

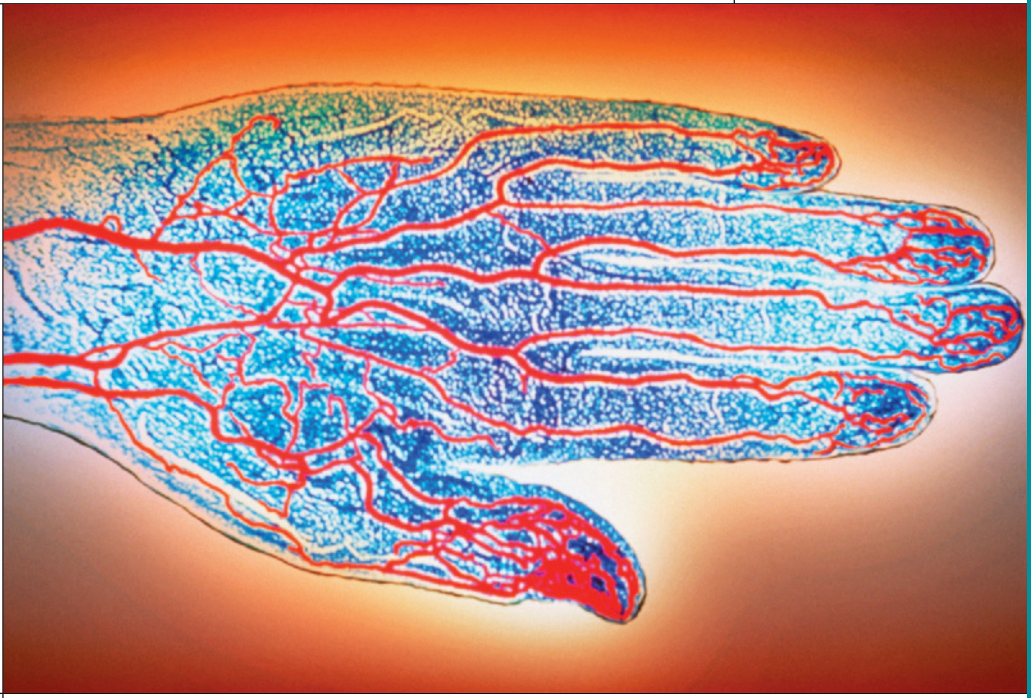
4. Use your answer to Problem 2 to show that α is the same under dose fractionation as it is for the single dose case in Problem 1.

5. Using your answers from Problems 3 and 4, explain how it is possible to use dose fractionation to increase ρ without also increasing α.

One of the reasons different drug doses and interdose intervals are used for different infections is to achieve different values of α, ρ, and τ.

Applications of Integrals

6



Arteries of the human hand are shown in a colorized X ray. In Section 6.3 we use an integral to calculate the flux in an artery (the volume of blood that passes a cross-section of an artery per unit time).

GJLP / CNRI / SPL / Science Source

WE HAVE ALREADY SEEN some of the applications of integrals in Chapter 5: drug pharmacokinetics, measles pathogenesis, bacterial growth, basal metabolism, breathing cycles, dialysis treatment, tumor growth, photosynthesis, medical imaging, bioavailability, the sterile insect technique, the rumen microbial ecosystem, and the spread of drug use.

In this chapter we explore some additional applications of integration: areas between curves, cerebral blood flow, disease progresssion, average values, survival and renewal, cardiac output, and volumes.

6.1 Areas Between Curves

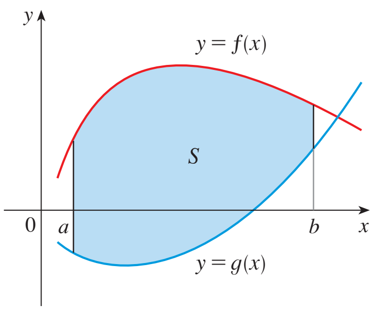


FIGURE 1
$S = \{(x, y) \mid a \leqslant x \leqslant b, g(x) \leqslant y \leqslant f(x)\}$

In Chapter 5 we defined and calculated areas of regions that lie under the graphs of functions. Here we use integrals to find areas of regions that lie between the graphs of two functions and then we show how this idea occurs in cerebral blood flow. As well, in the project after this section we see how the study of measles pathogenesis involves areas between curves.

Consider the region S that lies between two curves $y = f(x)$ and $y = g(x)$ and between the vertical lines $x = a$ and $x = b$, where f and g are continuous functions and $f(x) \geqslant g(x)$ for all x in $[a, b]$. (See Figure 1.)

Just as we did for areas under curves in Section 5.1, we divide S into n strips of equal width and then we approximate the ith strip by a rectangle with base Δx and height $f(x_i^*) - g(x_i^*)$. (See Figure 2. If we like, we could take all of the sample points to be right endpoints, in which case $x_i^* = x_i$.) The Riemann sum

$$\sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)]\,\Delta x$$

is therefore an approximation to what we intuitively think of as the area of S.

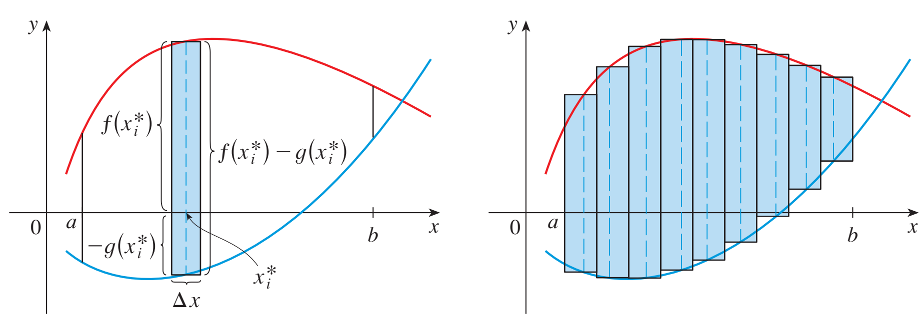


FIGURE 2 (a) Typical rectangle (b) Approximating rectangles

This approximation appears to become better and better as $n \to \infty$. Therefore we define the **area** A of the region S as the limiting value of the sum of the areas of these approximating rectangles.

(1)
$$\boxed{A = \lim_{n \to \infty} \sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)]\,\Delta x}$$

We recognize the limit in (1) as the definite integral of $f - g$. Therefore we have the following formula for area.

(2) The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f and g are continuous and $f(x) \geqslant g(x)$ for all x in $[a, b]$, is

$$A = \int_a^b [f(x) - g(x)]\,dx$$

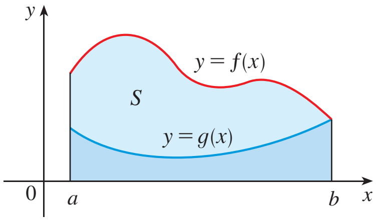


FIGURE 3

$A = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$

Notice that in the special case where $g(x) = 0$, S is the region under the graph of f and our general definition of area (1) reduces to our previous definition (Definition 5.1.2).

In the case where both f and g are positive, you can see from Figure 3 why (2) is true:

$$\begin{aligned} A &= [\text{area under } y = f(x)] - [\text{area under } y = g(x)] \\ &= \int_a^b f(x)\,dx - \int_a^b g(x)\,dx = \int_a^b [f(x) - g(x)]\,dx \end{aligned}$$

EXAMPLE 1 | Find the area of the region bounded above by $y = e^x$, bounded below by $y = x$, and bounded on the sides by $x = 0$ and $x = 1$.

SOLUTION The region is shown in Figure 4. The upper boundary curve is $y = e^x$ and the lower boundary curve is $y = x$. So we use the area formula (2) with $f(x) = e^x$, $g(x) = x$, $a = 0$, and $b = 1$:

$$\begin{aligned} A &= \int_0^1 (e^x - x)\,dx = e^x - \tfrac{1}{2}x^2\Big]_0^1 \\ &= e - \tfrac{1}{2} - 1 = e - 1.5 \end{aligned}$$ ■

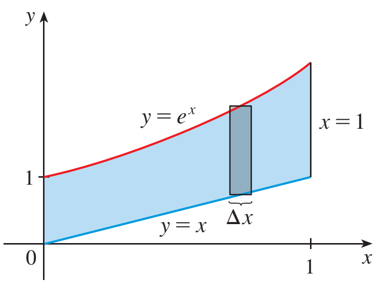


FIGURE 4

In Figure 4 we drew a typical approximating rectangle with width Δx as a reminder of the procedure by which the area is defined in (1). In general, when we set up an integral for an area, it's helpful to sketch the region to identify the top curve y_T, the bottom curve y_B, and a typical approximating rectangle as in Figure 5. Then the area of a typical rectangle is $(y_T - y_B)\,\Delta x$ and the equation

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} (y_T - y_B)\,\Delta x = \int_a^b (y_T - y_B)\,dx$$

summarizes the procedure of adding (in a limiting sense) the areas of all the typical rectangles.

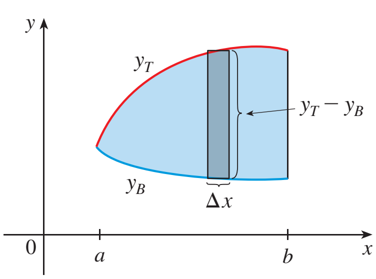


FIGURE 5

Notice that in Figure 5 the left-hand boundary reduces to a point, whereas in Figure 3 the right-hand boundary reduces to a point. In the next example both of the side boundaries reduce to a point, so the first step is to find a and b.

EXAMPLE 2 | Find the area of the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$.

SOLUTION We first find the points of intersection of the parabolas by solving their equations simultaneously. This gives $x^2 = 2x - x^2$, or $2x^2 - 2x = 0$. Thus $2x(x - 1) = 0$, so $x = 0$ or 1. The points of intersection are $(0, 0)$ and $(1, 1)$.

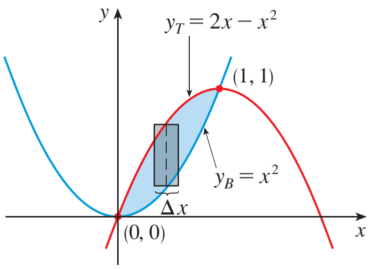


FIGURE 6

We see from Figure 6 that the top and bottom boundaries are

$$y_T = 2x - x^2 \qquad \text{and} \qquad y_B = x^2$$

The area of a typical rectangle is

$$(y_T - y_B)\,\Delta x = (2x - x^2 - x^2)\,\Delta x = (2x - 2x^2)\,\Delta x$$

and the region lies between $x = 0$ and $x = 1$. So the total area is

$$A = \int_0^1 (2x - 2x^2)\,dx = 2\int_0^1 (x - x^2)\,dx$$

$$= 2\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}$$

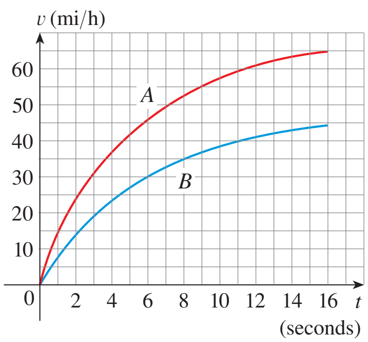


FIGURE 7

EXAMPLE 3 | **Interpreting the area between velocity curves** Figure 7 shows velocity curves for two cars, A and B, that start side by side and move along the same road. What does the area between the curves represent? Use the Midpoint Rule to estimate it.

SOLUTION We know from Section 5.3 that the area under the velocity curve A represents the distance traveled by car A during the first 16 seconds. Similarly, the area under curve B is the distance traveled by car B during that time period. So the area between these curves, which is the difference of the areas under the curves, is the distance between the cars after 16 seconds. We read the velocities from the graph and convert them to feet per second ($1 \text{ mi/h} = \frac{5280}{3600} \text{ ft/s}$).

t	0	2	4	6	8	10	12	14	16
v_A	0	34	54	67	76	84	89	92	95
v_B	0	21	34	44	51	56	60	63	65
$v_A - v_B$	0	13	20	23	25	28	29	29	30

We use the Midpoint Rule with $n = 4$ intervals, so that $\Delta t = 4$. The midpoints of the intervals are $\bar{t}_1 = 2$, $\bar{t}_2 = 6$, $\bar{t}_3 = 10$, and $\bar{t}_4 = 14$. We estimate the distance between the cars after 16 seconds as follows:

$$\int_0^{16} (v_A - v_B)\,dt \approx \Delta t\,[13 + 23 + 28 + 29]$$

$$= 4(93) = 372 \text{ ft}$$

Cerebral Blood Flow

In a paper[1] published in 1948, Seymour Kety and Carl Schmidt described a method for measuring cerebral blood flow in which the patient inhales a mixture of gases including a tracer of 15% nitrous oxide. Let $A(t)$ be the arterial concentration of N_2O measured as blood enters the brain and $V(t)$ the venous concentration of N_2O in blood flowing out of the brain in the jugular vein. Figure 8 shows typical graphs of $A(t)$ and $V(t)$, which are measured in units of mL of N_2O per mL of blood. Although $A(t) > V(t)$, you can see

1. S. Kety et al., "The Nitrous Oxide Method for the Quantitative Determination of Cerebral Blood Flow in Man: Theory, Procedure and Normal Values," *Journal of Clinical Investigation* 27 (1948): 476–83.

that after about 10 minutes $A(t)$ and $V(t)$ are almost the same because the brain is becoming saturated with nitrous oxide.

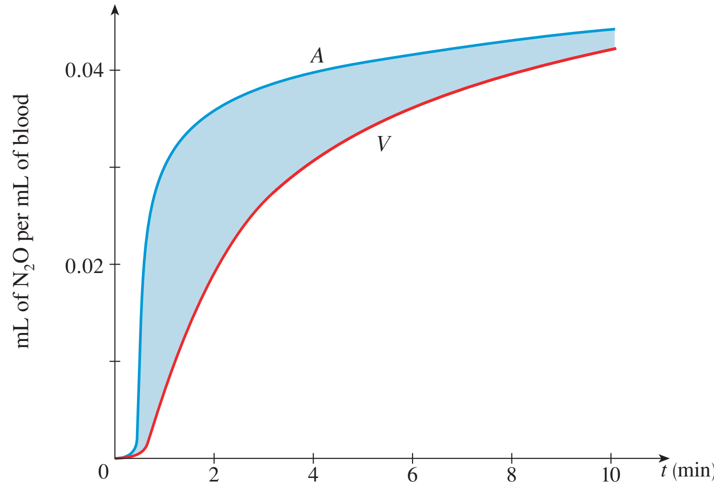


FIGURE 8
Arterial and venous concentrations of N_2O

The area between the curves A and V, shaded in Figure 8, plays a key role in calculating the **cerebral blood flow** F, which we measure in mL/min. We first consider $\int_0^{10} A(t)\,dt$. If we divide the interval $[0, 10]$ into subintervals of equal length Δt, then the volume of N_2O that flows past a point in the artery during such a subinterval from $t = t_{i-1}$ to $t = t_i$ is approximately

$$(\text{concentration of } N_2O \text{ in blood}) \cdot (\text{volume of blood}) = A(t_i)(F\,\Delta t)$$

Assuming that F remains constant, we see that the total volume of N_2O that enters the brain during the first 10 minutes is approximately

$$\sum_{i=1}^{n} A(t_i)\,F\,\Delta t = F \sum_{i=1}^{n} A(t_i)\,\Delta t$$

If we now let $n \to \infty$, we get the total quantity of N_2O brought to the brain during the first 10 minutes:

$$F \int_0^{10} A(t)\,dt$$

A similar argument shows that the quantity of N_2O that leaves the brain during this time period is

$$F \int_0^{10} V(t)\,dt$$

The difference of these quantities

$$F \int_0^{10} [A(t) - V(t)]\,dt$$

is therefore the quantity of N_2O that is taken up by the whole brain during the 10 minutes of inhalation. Let's call this quantity $Q_B(10)$. Then

$$Q_B(10) = F \int_0^{10} [A(t) - V(t)]\,dt$$

and therefore

(3) $$F = \frac{Q_B(10)}{\int_0^{10} [A(t) - V(t)]\, dt}$$

It turns out that $Q_B(10)$ can be found by other methods, so if the area between the curves is known, Equation 3 can then be used to calculate the cerebral blood flow.

EXAMPLE 4 | Cerebral blood flow

(a) Use the Midpoint Rule with five subintervals to estimate the area between the curves A and V in Figure 8.
(b) If it is known that the amount of N_2O absorbed by the brain is $Q_B(10) = 60$ mL, determine the cerebral blood flow.

SOLUTION

(a) We divide the interval $[0, 10]$ into five subintervals, whose midpoints are $t = 1, 3, 5, 7$, and 9. Then we use the graphs in Figure 8 to estimate the values of $A(t)$ and $V(t)$ at these midpoints and calculate their differences:

t	$A(t)$	$V(t)$	$A(t) - V(t)$
1	0.029	0.007	0.022
3	0.038	0.027	0.011
5	0.040	0.033	0.007
7	0.042	0.037	0.005
9	0.043	0.041	0.002

Using the Midpoint Rule with $\Delta t = 2$ min, we get the following estimate:

$$\int_0^{10} [A(t) - V(t)]\, dt \approx 2[0.022 + 0.011 + 0.007 + 0.005 + 0.002] = 0.094$$

So the area between the curves A and V is approximately 0.094 (mL/mL) · min.

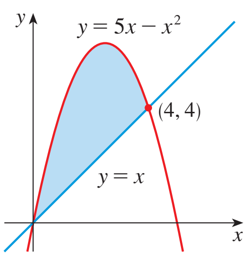
(b) With $Q_B(10) = 60$ mL and our result from part (a), we use Equation 3 to calculate the cerebral blood flow:

$$F = \frac{Q_B(10)}{\int_0^{10} [A(t) - V(t)]\, dt} \approx \frac{60}{0.094} \approx 640 \text{ mL/min}$$ ■

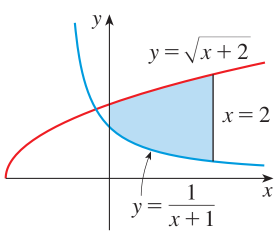
EXERCISES 6.1

1–2 Find the area of the shaded region.

1.



2.



3–6 Sketch the region enclosed by the given curves. Draw a typical approximating rectangle and label its height and width. Then find the area of the region.

3. $y = e^x, \quad y = x^2 - 1, \quad x = -1, \quad x = 1$

4. $y = \ln x, \quad xy = 4, \quad x = 1, \quad x = 3$

5. $y = x^2, \quad y^2 = x$

6. $y = x^2 - 2x, \quad y = x + 4$

7–10 Sketch the region enclosed by the given curves and find its area.

7. $y = 12 - x^2, \quad y = x^2 - 6$

8. $y = x^2, \quad y = 4x - x^2$

9. $y = e^x, \quad y = xe^x, \quad x = 0$

10. $y = \cos x, \quad y = 2 - \cos x, \quad 0 \leqslant x \leqslant 2\pi$

11–12 Use a graph to find approximate x-coordinates of the points of intersection of the given curves. Then find (approximately) the area of the region bounded by the curves.

11. $y = x\sin(x^2), \quad y = x^4$

12. $y = x\cos x, \quad y = x^{10}$

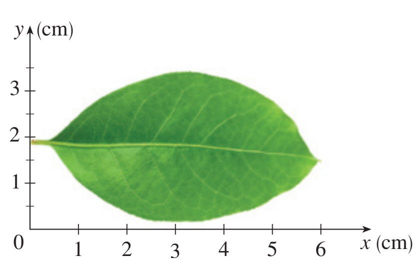
13. Sketch the region that lies between the curves $y = \cos x$ and $y = \sin 2x$ and between $x = 0$ and $x = \pi/2$. Notice that the region consists of two separate parts. Find the area of this region.

14. Sketch the curves $y = \cos x$ and $y = 1 - \cos x$, $0 \leqslant x \leqslant \pi$, and observe that the region between them consists of two separate parts. Find the area of this region.

15. Sometimes it's easier to find an area by regarding x as a function of y instead of y as a function of x. To illustrate this idea, let S be the region enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.
(a) By sketching S, observe that if you want to integrate with respect to x you have to split S into two parts with different boundary curves.
(b) If you integrate with respect to y, observe that there is a left boundary curve and a right boundary curve.
(c) Find the area of S using the method of either part (a) or part (b).

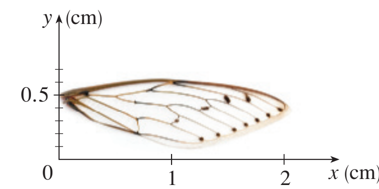
16. Find the area of the region enclosed by the curves $y = x$ and $4x + y^2 = 12$.

17. A **laurel leaf** is shown. Estimate its area using the Midpoint Rule with six subintervals.



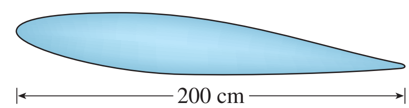
© Vasilius / Shutterstock.com

18. A **cicada wing** is shown. Estimate its area using the Midpoint Rule with six subintervals.

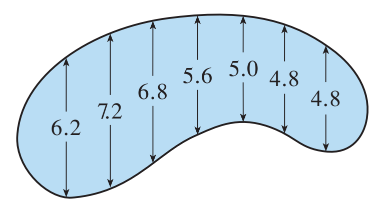


© Tropper2000 / Shutterstock.com

19. A cross-section of an airplane wing is shown. Measurements of the thickness of the wing, in centimeters, at 20-centimeter intervals are 5.8, 20.3, 26.7, 29.0, 27.6, 27.3, 23.8, 20.5, 15.1, 8.7, and 2.8. Use the Midpoint Rule to estimate the area of the wing's cross-section.



20. The widths (in meters) of a kidney-shaped swimming pool were measured at 2-meter intervals as indicated in the figure. Use the Midpoint Rule to estimate the area of the pool.



21. Cerebral blood flow The table shows measurements of $A(t)$, the concentration of N_2O flowing into a patient's brain, and $V(t)$, the concentration of N_2O flowing out of the brain, where t is measured in minutes and $A(t)$ and $V(t)$ are measured in mL of N_2O per mL of blood.

t	$A(t)$	$V(t)$
1	0.031	0.008
3	0.041	0.029
5	0.042	0.035
7	0.044	0.042
9	0.045	0.044

(a) Use the Midpoint Rule to estimate $\int_0^{10} [A(t) - V(t)]\, dt$.
(b) If the volume of N_2O absorbed by the brain in the first 10 minutes is 64 mL, calculate the cerebral blood flow.

22. Cerebral blood flow Models for the arterial and venous concentration functions in Figure 8 are given by

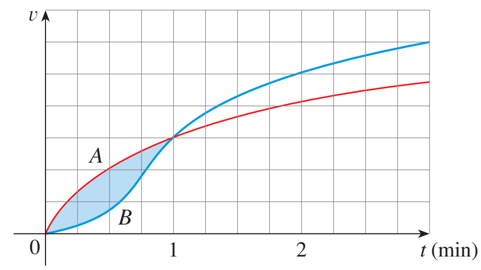
$$A(t) = \frac{0.05t^2}{t^2 + 1} \qquad V(t) = \frac{0.05t^2}{t^2 + 7}$$

(a) Find the area between the graphs of A and V for $0 \leqslant t \leqslant 10$.
(b) If the volume of N_2O absorbed by the brain in the first 10 minutes is 60 mL, determine the cerebral blood flow.

23. Racing cars driven by Chris and Kelly are side by side at the start of a race. The table shows the velocities of each car (in miles per hour) during the first 10 seconds of the race. Use the Midpoint Rule to estimate how much farther Kelly travels than Chris does during the first 10 seconds.

t	v_C	v_K	t	v_C	v_K
0	0	0	6	69	80
1	20	22	7	75	86
2	32	37	8	81	93
3	46	52	9	86	98
4	54	61	10	90	102
5	62	71			

24. Two cars, A and B, start side by side and accelerate from rest. The figure shows the graphs of their velocity functions.
(a) Which car is ahead after one minute? Explain.
(b) What is the meaning of the area of the shaded region?
(c) Which car is ahead after two minutes? Explain.
(d) Estimate the time at which the cars are again side by side.



25. Birth and death rates If the birth rate of a population is $b(t) = 2200e^{0.024t}$ people per year and the death rate is $d(t) = 1460e^{0.018t}$ people per year, find the area between these curves for $0 \leqslant t \leqslant 10$. What does this area represent?

26. Find the number a such that the line $x = a$ bisects the area under the curve $y = 1/x^2$, $1 \leqslant x \leqslant 4$.

27. Find the values of c such that the area of the region bounded by the parabolas $y = x^2 - c^2$ and $y = c^2 - x^2$ is 576.

28. Find the area of the region bounded by the parabola $y = x^2$, the tangent line to this parabola at (1, 1), and the x-axis.

PROJECT Disease Progression and Immunity

In Section 5.1 we considered the progression of measles in a patient with no immunity and graphed the measles pathogenesis curve $N = f(t)$ (page 325), which we modeled with the function

$$f(t) = -t(t - 21)(t + 1)$$

We saw that the total amount of infection by time t (measured by the area under the pathogenesis curve up to time t) played an important role in determining whether individuals develop symptoms. In particular, symptoms appear only after the total amount of infection exceeds 7848 (cells/mL) $\times$ days, which occurs at day 12 of the infection for individuals with no immunity, that is,

$$\int_0^{12} f(t)\, dt = 7848 \text{ (cells/mL)} \times \text{days}$$

1. Plot the curves $cf(t)$ for $c = 0.9, 0.85, 0.8, 0.6$, and 0.4. These resemble curves for patients that have increasing levels of immunity against the virus at the time of the infection.

2. Some of the patients in Problem 1 will develop symptoms and some will not. Find the areas under the curve from $t = 0$ to $t = 21$ for each value of c and compare them with the value 7848 that is needed to display symptoms. Which patients will become symptomatic at some point during their infection?

The term *infectiousness* refers to the extent to which the disease is transmitted between individuals. For patients without immunity, we saw in Section 5.1 that infectiousness begins around day $t_1 = 10$ and ends around day $t_2 = 18$. Infectiousness begins on day 10 because the concentration of virus in the plasma after 10 days [that is, the value $f(10)$] is the threshold concentration required before any transmission can occur. Further, infectiousness ends on day 18 because the immune system manages to prevent further transmission from this point onward.

3. Plot the points $P_1 = (t_1, f(t_1))$ and $P_2 = (t_2, f(t_2))$ on the graph of f. These points show the values of f at the beginning and end of the infectious period. Draw a line between the points. What is the slope of this line? Find an equation of this line.

4. Given that $L = f(t_1)$ is the threshold concentration of the virus required for transmission to begin, plot the point P_3 on the curve $N = 0.9f(t)$ where $0.9f(t) = L$. The value of t satisfying this equation is the time at which infectiousness begins for a patient with $c = 0.9$. It has been shown that the time at which infectiousness ends for such patients can again be determined by drawing a line through P_3 with the same slope as that in Problem 3 and then determining the time t_4 at which it intersects the curve $N = 0.9f(t)$. Draw this line on the graph and determine t_4.

5. Repeat Problem 4 for $cf(t)$ with $c = 0.85$, 0.8, 0.6, and 0.4. Note that some patients may not have a point corresponding to P_3.

6. Find the area between the graph of f and the line in Problem 3. This area represents the total level of infectiousness of an infected person. (See Figure 1.)

7. Find the areas enclosed by the curves $N = cf(t)$ and their corresponding intersecting lines for $c = 0.9$, 0.85, 0.8, 0.6, and 0.4. (Note that this will not be feasible for patients that have no infectious period.) Compare these areas to the one found in Problem 6.)

8. Which patients are
(a) symptomatic and infectious?
(b) symptomatic and noninfectious?
(c) asymptomatic and noninfectious?

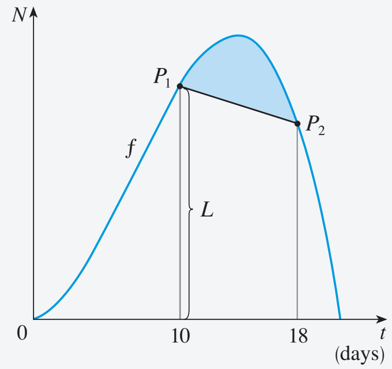


FIGURE 1

■ PROJECT The Gini Index

How is it possible to measure the distribution of income among the inhabitants of a given country? One such measure is the *Gini index*, named after the Italian economist Corrado Gini, who first devised the index in 1912.

We first rank all households in a country by income and then we compute the percentage of households whose income is at most a given percentage of the country's total income. We define a **Lorenz curve** $y = L(x)$ on the interval $[0, 1]$ by plotting the point $(a/100, b/100)$ on the curve if the bottom $a\%$ of households receive at most $b\%$ of the

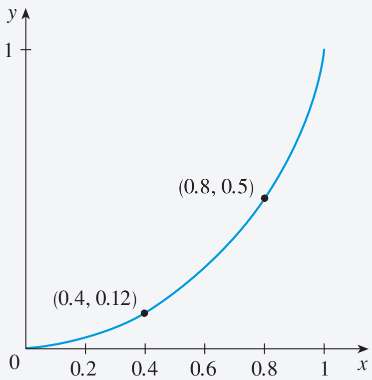


FIGURE 1
Lorenz curve for the United States in 2010

total income. For instance, in Figure 1 the point (0.4, 0.12) is on the Lorenz curve for the United States in 2010 because the poorest 40% of the population received just 12% of the total income. Likewise, the bottom 80% of the population received 50% of the total income, so the point (0.8, 0.5) lies on the Lorenz curve. (The Lorenz curve is named after the American economist Max Lorenz.)

Figure 2 shows some typical Lorenz curves. They all pass through the points (0, 0) and (1, 1) and are concave upward. In the extreme case $L(x) = x$, society is perfectly egalitarian: The poorest a% of the population receives a% of the total income and so everybody receives the same income. The area between a Lorenz curve $y = L(x)$ and the line $y = x$ measures how much the income distribution differs from absolute equality. The **Gini index** (sometimes called the **Gini coefficient** or the **coefficient of inequality**) is the area between the Lorenz curve and the line $y = x$ (shaded in Figure 3) divided by the area under $y = x$.

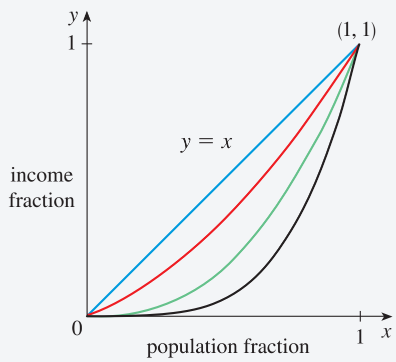


FIGURE 2

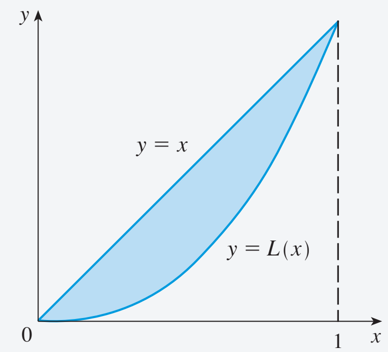


FIGURE 3

1. (a) Show that the Gini index G is twice the area between the Lorenz curve and the line $y = x$, that is,

$$G = 2\int_0^1 [x - L(x)]\,dx$$

(b) What is the value of G for a perfectly egalitarian society (everybody has the same income)? What is the value of G for a perfectly totalitarian society (a single person receives all the income)?

2. The following table (derived from data supplied by the US Census Bureau) shows values of the Lorenz function for income distribution in the United States for the year 2010.

x	0.0	0.2	0.4	0.6	0.8	1.0
$L(x)$	0.000	0.034	0.120	0.266	0.498	1.000

(a) What percentage of the total US income was received by the richest 20% of the population in 2010?

(b) Use a calculator or computer to fit a quadratic function to the data in the table. Graph the data points and the quadratic function. Is the quadratic model a reasonable fit?

(c) Use the quadratic model for the Lorenz function to estimate the Gini index for the United States in 2010.

3. The following table gives values for the Lorenz function in the years 1970, 1980, 1990, and 2000. Use the method of Problem 2 to estimate the Gini index for the United States for those years and compare with your answer to Problem 2(c). Do you notice a trend?

x	0.0	0.2	0.4	0.6	0.8	1.0
1970	0.000	0.041	0.149	0.323	0.568	1.000
1980	0.000	0.042	0.144	0.312	0.559	1.000
1990	0.000	0.038	0.134	0.293	0.530	1.000
2000	0.000	0.036	0.125	0.273	0.503	1.000

CAS **4.** A power model often provides a more accurate fit than a quadratic model for a Lorenz function. If you have a computer with Maple or Mathematica, fit a power function ($y = ax^k$) to the data in Problem 2 and use it to estimate the Gini index for the United States in 2010. Compare with your answer to parts (b) and (c) of Problem 2.

6.2 Average Values

It is easy to calculate the average value of finitely many numbers $y_1, y_2, \ldots, y_n$:

$$y_{\text{ave}} = \frac{y_1 + y_2 + \cdots + y_n}{n}$$

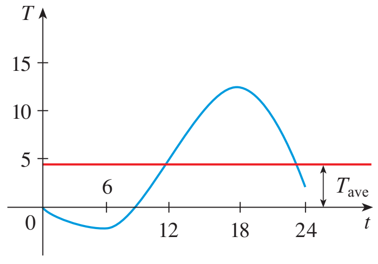


FIGURE 1

But how do we compute the average temperature during a day if infinitely many temperature readings are possible? Figure 1 shows the graph of a temperature function $T(t)$, where t is measured in hours and T in °C, and a guess at the average temperature, T_{ave}.

In general, let's try to compute the average value of a function $y = f(x)$, $a \leqslant x \leqslant b$. We start by dividing the interval $[a, b]$ into n equal subintervals, each with length $\Delta x = (b - a)/n$. Then we choose points $x_1^*, \ldots, x_n^*$ in successive subintervals and calculate the average of the numbers $f(x_1^*), \ldots, f(x_n^*)$:

$$\frac{f(x_1^*) + \cdots + f(x_n^*)}{n}$$

(For example, if f represents a temperature function and $n = 24$, this means that we take temperature readings every hour and then average them.) Since $\Delta x = (b - a)/n$, we can write $n = (b - a)/\Delta x$ and the average value becomes

$$\frac{f(x_1^*) + \cdots + f(x_n^*)}{\dfrac{b-a}{\Delta x}} = \frac{1}{b-a}\left[f(x_1^*)\,\Delta x + \cdots + f(x_n^*)\,\Delta x\right]$$

$$= \frac{1}{b-a}\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

If we let n increase, we would be computing the average value of a large number of closely spaced values. (For example, we would be averaging temperature readings taken every minute or even every second.) The limiting value is

$$\lim_{n\to\infty} \frac{1}{b-a} \sum_{i=1}^{n} f(x_i^*)\,\Delta x = \frac{1}{b-a}\int_a^b f(x)\,dx$$

by the definition of a definite integral.

Therefore we define the **average value of f** on the interval $[a, b]$ as

For a positive function, we can think of this definition as saying

$$\frac{\text{area}}{\text{width}} = \text{average height}$$

$$\boxed{f_{\text{ave}} = \frac{1}{b-a}\int_a^b f(x)\,dx}$$

EXAMPLE 1 | Find the average value of the function $f(x) = 1 + x^2$ on the interval $[-1, 2]$.

SOLUTION With $a = -1$ and $b = 2$ we have

$$\begin{aligned} f_{\text{ave}} &= \frac{1}{b-a}\int_a^b f(x)\,dx = \frac{1}{2-(-1)}\int_{-1}^{2} (1 + x^2)\,dx \\ &= \frac{1}{3}\left[x + \frac{x^3}{3}\right]_{-1}^{2} = 2 \end{aligned}$$

■

EXAMPLE 2 | World population In Chapter 1 we modeled the size of the human population of the world with the exponential function

$$P(t) = (1.43653 \times 10^9) \cdot (1.01395)^t$$

where t is measured in years and $t = 0$ corresponds to the year 1900. What was the average population in the 20th century?

SOLUTION The average population for $0 \leqslant t \leqslant 100$ was

$$\begin{aligned} P_{\text{ave}} &= \frac{1}{100}\int_0^{100} (1.43653 \times 10^9) \cdot (1.01395)^t\,dt \\ &= \frac{1}{100} \cdot (1.43653 \times 10^9) \cdot \left[\frac{(1.01395)^t}{\ln(1.01395)}\right]_{t=0}^{t=100} \\ &= 1.43653 \times 10^7 \cdot \frac{(1.01395)^{100} - 1}{\ln(1.01395)} \\ &\approx 310 \times 10^7 = 3.1 \times 10^9 \end{aligned}$$

So the average world population in the 20th century was about 3.1 billion. ■

If $T(t)$ is the temperature at time t, we might wonder if there is a specific time when the temperature is the same as the average temperature. For the temperature function graphed in Figure 1, we see that there are two such times—just before noon and just before midnight. In general, is there a number c at which the value of a function f is exactly equal to the average value of the function, that is, $f(c) = f_{\text{ave}}$? The following theorem says that this is true for continuous functions.

The Mean Value Theorem for Integrals If f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that

$$f(c) = f_{\text{ave}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

that is,
$$\int_a^b f(x)\,dx = f(c)(b-a)$$

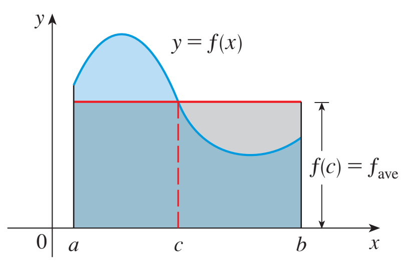


FIGURE 2

You can always chop off the top of a (two-dimensional) mountain at a certain height and use it to fill in the valleys so that the mountain becomes completely flat.

The Mean Value Theorem for Integrals is a consequence of the Mean Value Theorem for derivatives and the Fundamental Theorem of Calculus. The proof is outlined in Exercise 23.

The geometric interpretation of the Mean Value Theorem for Integrals is that, for *positive* functions f, there is a number c such that the rectangle with base $[a, b]$ and height $f(c)$ has the same area as the region under the graph of f from a to b. (See Figure 2 and the more picturesque interpretation in the margin note.)

EXAMPLE 3 | Since $f(x) = 1 + x^2$ is continuous on the interval $[-1, 2]$, the Mean Value Theorem for Integrals says there is a number c in $[-1, 2]$ such that

$$\int_{-1}^{2} (1 + x^2)\,dx = f(c)[2 - (-1)]$$

In this particular case we can find c explicitly. From Example 1 we know that $f_{\text{ave}} = 2$, so the value of c satisfies

$$f(c) = f_{\text{ave}} = 2$$

Therefore $\quad 1 + c^2 = 2 \quad$ so $\quad c^2 = 1$

So in this case there happen to be two numbers $c = \pm 1$ in the interval $[-1, 2]$ that work in the Mean Value Theorem for Integrals. ■

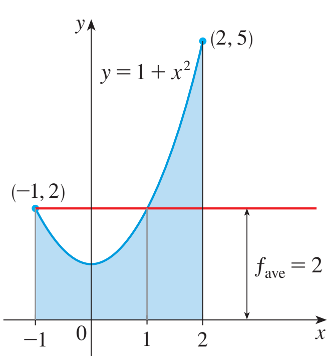


FIGURE 3

Examples 1 and 3 are illustrated by Figure 3.

EXERCISES 6.2

1–6 Find the average value of the function on the given interval.

1. $f(x) = 4x - x^2$, $[0, 4]$

2. $f(x) = \sin 4x$, $[-\pi, \pi]$

3. $g(x) = \sqrt[3]{x}$, $[1, 8]$

4. $f(\theta) = \sec^2(\theta/2)$, $[0, \pi/2]$

5. $h(x) = \cos^4 x \sin x$, $[0, \pi]$

6. $h(u) = (3 - 2u)^{-1}$, $[-1, 1]$

7–10
(a) Find the average value of f on the given interval.
(b) Find c such that $f_{\text{ave}} = f(c)$.
(c) Sketch the graph of f and a rectangle whose area is the same as the area under the graph of f.

7. $f(x) = (x - 3)^2$, $[2, 5]$

8. $f(x) = \ln x$, $[1, 3]$

9. $f(x) = 2\sin x - \sin 2x$, $[0, \pi]$

10. $f(x) = 2x/(1 + x^2)^2$, $[0, 2]$

11. Find the average value of f on $[0, 8]$.